HSPT Exam Simulator

Full-Length Practice Exam 09

Complete printable test with 298 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Verbal Skills	60	20 minutes
Quantitative Skills	52	35 minutes
Reading Comprehension	62	35 minutes
Mathematics	64	45 minutes
Language	60	30 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 165 minutes. Calculators should be used only when permitted by the rules followed for your HSPT preparation.

Section 1: Verbal Skills

60 questions | Recommended time: 20 minutes

1. The host prepared a LAVISH feast, with more dishes than the guests could count. LAVISH most nearly means:

A) barely sufficient

B) hastily arranged

C) carefully concealed

D) extravagantly abundant

2. The puppet had a WINSOME expression that immediately charmed the audience. WINSOME most nearly means:

A) sternly commanding

B) openly suspicious

C) appealingly charming

D) deliberately boastful

3. The BENIGN remark was intended to encourage, not to wound. BENIGN most nearly means:

A) cleverly insulting

B) strictly factual

C) deliberately puzzling

D) kindly and harmless

4. The abandoned settlement looked DESOLATE, with empty streets and shuttered homes. Which word most nearly expresses the opposite of DESOLATE here?

A) bustling

B) deserted

C) forlorn

D) uninhabited

5. The SPRIGHTLY dancer moved with energy and lightness. Which word most nearly expresses the opposite of SPRIGHTLY?

A) lively

B) animated

C) lethargic

D) vivacious

6. The RECALCITRANT trainee resisted even reasonable instructions. Which word most nearly expresses the opposite of RECALCITRANT?

A) amenable

B) obstinate

C) uncooperative

D) intractable

7. A tailor leaves extra cloth beyond a seam so the garment can later be adjusted. Which action preserves a similar reserve for later correction?

A) A painter mixes two colors to create a third.

B) A carpenter leaves a board slightly long so it can be trimmed to fit.

C) A porter labels a crate after it has been delivered.

D) A driver replaces a worn tire before a journey.

8. Increasing playback volume makes a recording sound louder without changing the stored recording. Which change has the most similar relationship to its source?

A) Cropping and saving an image removes part of the original file.

B) Zooming in on a digital image enlarges its display without editing the saved image.

C) Recording a new ending changes the content of a song.

D) Erasing a paragraph removes it from a document.

9. An interpreter begins translating a speech while the speaker is still delivering later sentences. Which process similarly overlaps producing later material with delivering earlier material?

A) A conveyor carries the first packed cartons away while later cartons are still being packed.

B) A courier waits until every carton is packed before collecting any.

C) An inspector rereads a report after the entire job is complete.

D) A clerk sorts old receipts after the shop closes.

10. Which word belongs with PLEDGE, VOW, and PROMISE as a commitment to future conduct?

A) undertaking

B) prediction

C) recollection

D) description

11. Which group contains ONLY accounts in which writers describe their own lives or experiences?

A) biography, weather bulletin, atlas

B) memoir, autobiography, personal diary

C) instruction manual, index, glossary

D) novel, dictionary, sales catalog

12. Which word names an arrangement into levels of rank rather than a mere collection of listed items?

A) inventory

B) checklist

C) hierarchy

D) catalog

13. The ordinary rule keeps every parcel sealed. A stated exception requires opening parcels marked INSPECT. No other label changes either rule. A parcel is marked URGENT but not INSPECT. What do the rules require?

A) Open the parcel because urgency creates an exception.

B) Keep the parcel sealed.

C) Remove its address before deciding.

D) Treat the URGENT label as if it said INSPECT.

14. Reports are sorted by author's surname first; titles are compared only when surnames match. Ames wrote Zebra and Willow; Bell wrote Acorn. Under these rules, which report comes first?

A) Acorn by Bell

B) Zebra by Ames

C) The rules cannot distinguish these reports.

D) Willow by Ames

15. Every visitor at a reception spoke with at least one guide. Which additional statement would guarantee that all visitors spoke with the SAME guide?

A) Every guide spoke with at least one visitor.

B) Some visitors spoke with several guides.

C) Exactly one guide spoke with visitors at the reception.

D) Each guide wore the same kind of badge.

16. The map includes a TACTILE border that readers can follow with their fingertips. TACTILE most nearly means:

A) related to hearing

B) related to touch

C) visible from far away

D) arranged by date

17. As the dry season continued, the stream began to DWINDLE. DWINDLE most nearly means:

A) suddenly overflow

B) remain unchanged

C) gradually diminish

D) change direction

18. Even after reading the instructions twice, Luis could not FATHOM the reason for the extra step. FATHOM most nearly means:

A) announce publicly

B) understand fully

C) reject immediately

D) describe briefly

19. The SHREWD negotiator quickly recognized an offer designed to mislead her. Which word most nearly expresses the opposite of SHREWD here?

A) gullible

B) astute

C) perceptive

D) discerning

20. The speaker gave an ELOQUENT account, expressing difficult ideas clearly and gracefully. Which word most nearly expresses the opposite of ELOQUENT?

A) expressive

B) inarticulate

C) fluent

D) persuasive

21. Their SPONTANEOUS celebration arose without any advance arrangement. Which word most nearly expresses the opposite of SPONTANEOUS?

A) unprompted

B) premeditated

C) impulsive

D) unplanned

22. A crossing guard gives traffic and pedestrians alternating turns through a shared crossing. Which arrangement solves a similar conflict over shared access?

A) A cleaner washes every carriage before departure.

B) A painter gives each station a different color.

C) A dispatcher gives trains traveling in opposite directions separate turns on a single-track section.

D) A conductor checks whether each passenger has a ticket.

23. Scaffolding supports work on a building until that temporary support is no longer needed. Which has the most similar relationship?

A) Training wheels stabilize a bicycle while its rider is learning to balance, then can be removed.

B) A permanent foundation continues supporting a completed building.

C) A locked container keeps an object inaccessible.

D) A decorative border changes an object's appearance.

24. A parcel count can reveal that a shipment has the wrong number of boxes without revealing WHICH box is wrong. Which check has the most similar limitation?

A) A numbered list identifies every box and its destination.

B) A page-by-page comparison highlights the exact missing sentence.

C) A tracking label states which individual parcel has arrived.

D) A total word count can reveal that a document is too short without identifying the missing sentence.

25. Which cooking method belongs with SIMMERING, POACHING, and BOILING as a method using water or water vapor to transfer heat?

A) broiling

B) toasting

C) steaming

D) grilling

26. Which word belongs with PROVISIONAL, TENTATIVE, and INTERIM as a description of something that is not yet final?

A) preliminary

B) definitive

C) conclusive

D) settled

27. Which word does NOT belong with DETOUR, BYPASS, and WORKAROUND as a way of getting around an obstacle?

A) alternative route

B) diversion

C) impediment

D) circumvention

28. Rowan stands immediately behind Suri in a line. Tari, who is neither Rowan nor Suri, leaves the line. Everyone else keeps the same relative order and closes any gap. What must remain true?

A) Suri is now first in line.

B) Rowan is now last in line.

C) Someone must stand between Rowan and Suri.

D) Rowan is immediately behind Suri.

29. A finite, nonempty group of runners finishes a race with no ties. A description claims that EVERY member finished before another member of that same group. Why is that description impossible?

A) Every runner must have finished before someone outside the group.

B) The first finisher must have finished after the last finisher.

C) A group cannot include runners from the same race.

D) The last finisher in the group did not finish before any other member of the group.

30. A document says that the owner signed one page and the artist signed another. It does not forbid one person from holding both roles. What can be concluded from these role descriptions alone?

A) The owner and artist must be different people.

B) Only the owner could have signed both pages.

C) At least one signature must have been forged.

D) They do not establish that different people signed the pages.

31. The DEFT repairer adjusted the tiny clasp without bending its delicate pin. DEFT most nearly means:

A) hesitant

B) careless

C) skillful

D) forceful

32. Her WAYWARD thoughts kept wandering away from the task. WAYWARD most nearly means:

A) difficult to direct

B) firmly focused

C) carefully organized

D) quickly completed

33. The TURBULENT meeting ended amid interruptions and arguments. TURBULENT most nearly means:

A) routine

B) private

C) disorderly

D) brief

34. Her GRACIOUS reply thanked even those who had criticized her. Which word most nearly expresses the opposite of GRACIOUS here?

A) civil

B) polite

C) courteous

D) discourteous

35. The PLIABLE strip bent around the curved frame without difficulty. Which word most nearly expresses the opposite of PLIABLE?

A) supple

B) flexible

C) bendable

D) rigid

36. The chamber's AUSTERE appearance matched its single bench and unadorned walls. Which word most nearly expresses the opposite of AUSTERE here?

A) unembellished

B) plain

C) ornate

D) spare

37. A stop on a sliding drawer prevents it from traveling beyond its intended end position. Which device similarly prevents an object from passing a fixed limit?

A) A pulley changes the direction of a pull.

B) A cushion softens a seat.

C) A knot at the end of a cord keeps the cord from slipping completely through a hole.

D) A polished surface reflects more light.

38. Perforations let a sheet separate along an intended weak line. Which design similarly directs a break toward a deliberately weaker part?

A) A damaged hinge weakens accidentally through neglect.

B) A thicker cable increases strength throughout its length.

C) A locked cover prevents a container from opening.

D) A breakaway link is made weaker than the main chain so that the link gives way first.

39. Whether a key fits depends on the relationship between that key and a particular lock. Which property similarly depends on a match between TWO items?

A) Whether two words rhyme depends on a match between their ending sounds.

B) Whether a stone is heavy describes the stone's mass.

C) Whether a cloth is red describes its color.

D) Whether a page is torn describes its condition.

40. Which word belongs with HEARTH, FORGE, and KILN as a place or chamber in which heat from a fire is maintained?

A) silo

B) cistern

C) granary

D) furnace

41. Which expression does NOT belong with UPSWING, UPTICK, and CLIMB as a description of movement to a higher level?

A) plateau

B) rise

C) leap upward

D) increase

42. Which pair uses the SAME word first as an object and then as an action performed with that object?

A) a singer; sing the song

B) a stamp; stamp the envelope

C) a red coat; a red roof

D) he ran; she ran

43. A copying task says: Copy the words inside the quotation marks; do not carry out the quoted instruction: "Close the shutter." Which action follows the copying task?

A) Close the actual shutter.

B) Write the words Close the shutter.

C) Write the words Open the shutter.

D) Leave the response blank.

44. At noon, every banner on the west wall was red. At one o’clock, a blue banner hangs there. Which additional fact would make these reports inconsistent?

A) The blue banner was already on that wall at noon and has not changed color.

B) A red banner was removed after noon.

C) The blue banner was added after noon.

D) The west wall is wider than the east wall.

45. A one-to-one word code uses a different code word for each English word; word order may change. DAX LUM means quiet harbor. LUM VEV means quiet morning. DAX ZEP means busy harbor. Which English pair is represented by VEV ZEP?

A) quiet and harbor

B) morning and busy

C) busy and harbor

D) quiet and morning

46. His JOCULAR reply turned the tense exchange into laughter. JOCULAR most nearly means:

A) solemn

B) joking

C) uncertain

D) accusatory

47. Her FLIPPANT answer treated the important concern as if it deserved no serious attention. FLIPPANT most nearly means:

A) carefully reasoned

B) deeply sympathetic

C) disrespectfully casual

D) formally respectful

48. The SAGACIOUS adviser recognized a consequence that everyone else had overlooked. SAGACIOUS most nearly means:

A) wealthy and famous

B) cautious and silent

C) eager and impatient

D) wise and perceptive

49. The rope remained TAUT between the posts. Which word most nearly expresses the opposite of TAUT?

A) tight

B) stretched

C) tense

D) slack

50. The TRANSIENT stain disappeared after a few minutes. Which word most nearly expresses the opposite of TRANSIENT?

A) enduring

B) brief

C) fleeting

D) temporary

51. His MEEK response offered no resistance to the unfair demand. Which word most nearly expresses the opposite of MEEK here?

A) submissive

B) assertive

C) unresisting

D) timid

52. A gasket fills a narrow gap so liquid cannot escape between joined parts. Which action serves a comparable sealing purpose?

A) A handle makes a box easier to carry.

B) A hinge lets a lid move.

C) Caulking closes gaps around a window frame.

D) A label identifies a bottle’s contents.

53. A dim stage light can establish a background against which a small bright object stands out. Which arrangement uses a similar contrast?

A) A lid keeps dust away from a container.

B) A pale border makes dark lettering easier to distinguish.

C) A motor moves a belt at a steady speed.

D) A tray gathers several tools in one place.

54. One mismatching letter can prove that two texts are not identical, but one matching letter cannot prove that they are identical. Which comparison follows the same reasoning?

A) A list of dates can put events in chronological order.

B) One unequal pair disproves that all rods have equal length, but one equal pair does not establish it.

C) A label can distinguish one address from another.

D) A receipt can record when a payment was made.

55. Which word belongs with DAWN, DUSK, and NOON as a time of day?

A) horizon

B) compass

C) weather

D) midnight

56. Which word does NOT belong with FRAGRANCE, AROMA, and SCENT as a description of smell?

A) odor

B) perfume

C) bouquet

D) texture

57. Which group contains ONLY words naming protective outer coverings for objects?

A) blade, point, edge

B) echo, whisper, murmur

C) sheath, case, sleeve

D) hinge, axle, pivot

58. A rule applies only to ORIGINAL documents. A photograph of a document is a copy, even when perfectly clear. Which item is outside the rule's stated scope?

A) An original document with faded ink

B) An original document kept in a folder

C) A clear photograph of an original document

D) An original document signed yesterday

59. A map key says every dotted route is a footpath. A route is drawn as a solid line. Which conclusion follows from that rule alone?

A) The rule does not determine whether the solid route is a footpath.

B) The solid route cannot be a footpath.

C) Every footpath must be dotted.

D) The solid route must be a road.

60. Each of three machines stamps either ROUND or SQUARE on a card. A checker reports only whether all stamps agree. The report says they do NOT all agree. What is necessarily true?

A) At least one ROUND stamp and at least one SQUARE stamp appear.

B) Exactly one machine failed to stamp.

C) All three stamps are ROUND.

D) The first machine necessarily stamped SQUARE.

Section 2: Quantitative Skills

52 questions | Recommended time: 35 minutes

61. To obtain each next term, multiply the preceding term by its NUMBER of positive divisors, including 1 and the term itself. What comes next? 3, 6, 24, 192, ____

A) 3,072

B) 1,536

C) 2,688

D) 2,304

62. For n = 1, 2, 3, and so on, write the remainder from n ÷ 2 PLUS the remainder from n ÷ 3. The outputs begin 2, 2, 1, 1, 3, 0. What are the next TWO outputs?

A) 1, 1

B) 2, 1

C) 0, 3

D) 2, 2

63. In each next fraction, the old denominator becomes the numerator, and the sum of the old numerator and denominator becomes the denominator. What comes next? 1/3, 3/4, 4/7, 7/11, ____

A) 11/14

B) 11/18

C) 7/18

D) 18/11

64. A counter uses only the digits 0, 1, and 2. To increase it by one, change a final 0 to 1 or 1 to 2; a final 2 becomes 0 and carries one place left. What follows 12, 20, 21, 22, 100, ____?

A) 110

B) 101

C) 102

D) 200

65. A fraction with positive integer numerator and denominator becomes 1/2 when 1 is added only to its numerator. It becomes 1/3 when 1 is added only to its denominator. What is the original fraction?

A) 2/6

B) 4/10

C) 3/8

D) 3/9

66. Use each digit 1, 3, 6, and 8 exactly once to form two two-digit numbers whose sum is 99. What is the greatest possible product?

A) 2,318

B) 1,118

C) 1,328

D) 2,108

67. What is the least positive integer k such that 18 divides 12nk for EVERY positive integer n?

A) 2

B) 6

C) 9

D) 3

68. A nonempty row of seats alternates between occupied and empty seats, beginning and ending with occupied seats.
Quantity A: The number of occupied seats
Quantity B: The number of empty seats

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

69. Three consecutive positive integers have middle value n.
Quantity A: Their product
Quantity B: n^3

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

70. The numbers satisfy $0 < x < y$.
Quantity A: x/y
Quantity B: $(y - x)/y$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

71. A cylinder's curved side is cut along a vertical line and unrolled into a rectangle. The cylinder has radius r and height h.
Quantity A: The perimeter of the resulting rectangle
Quantity B: $4\pi r + 2h$

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

72. Each corner of a triangle is cut off by joining the midpoints of the two sides meeting there, leaving the central inverted triangle.
Quantity A: The perimeter of the remaining triangle
Quantity B: The perimeter of the original triangle

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

73. A convex polygon has at least five sides. At every vertex, one exterior angle supplements the interior angle.
Quantity A: The sum of its interior angles
Quantity B: The sum of those exterior angles

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

74. Begin with 2. Each next term is formed by appending the NUMBER OF DIGITS in the current term to its right. What comes next? 2, 21, 212, 2123, ____

A) 21233

B) 21234

C) 21232

D) 21235

75. List the positive integers that are divisible by NEITHER 2 NOR 3, in increasing order. What are the next TWO terms? 1, 5, 7, 11, 13, 17, 19, ____, ____

A) 23, 29

B) 21, 23

C) 23, 25

D) 25, 29

76. Begin with 4, 9, 2. Alternate these rules for new terms: first use the RANGE of the preceding THREE terms, then their MEDIAN, then range again, and so on. Range is greatest minus least; median is the middle value after sorting. What are the next TWO terms? 4, 9, 2, 7, 7, 5, ____, ____

A) 7, 2

B) 5, 2

C) 7, 5

D) 2, 7

77. A till rounds each cash transaction to the nearest $0.05. Two items each cost $1.02. How does paying for them TOGETHER compare with paying for them in two separate transactions?

A) Together costs $0.05 less.

B) The rounded totals are equal.

C) Together costs $0.10 more.

D) Together costs $0.05 more.

78. For $a \neq b$, define $a \square b = (a + b)/(a - b)$. If $a \square b = 3$, what is $b \square a$?

A) −3

B) 3

C) 1/3

D) −1/3

79. The integers a and b satisfy $2 < a < b$ and $1/2 + 1/a + 1/b = 1$. What is the ordered pair (a, b)?

A) (3, 4)

B) (4, 8)

C) (3, 6)

D) (4, 4)

80. How many integers from 1 through 60 are divisible by EXACTLY TWO of the numbers 2, 3, and 5?

A) 20

B) 14

C) 16

D) 12

81. The two-digit integer N has tens digit a and ones digit b.
Quantity A: N − (a + b)
Quantity B: 9a

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

82. The positive integers a and b satisfy a < b, and a/b is NOT in lowest terms.
Quantity A: The greatest common factor of a and b
Quantity B: The numerator when a/b is reduced to lowest terms

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

83. The integer n is at least 2.
Quantity A: 1/n − 1/(n + 1)
Quantity B: 1/(n(n − 1))

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

84. Three distinct points A, B, and C lie on a circle.
Quantity A: The length AB + BC, using straight segments
Quantity B: The length AC, using a straight segment

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

85. The vertices of a rectangle are A, B, C, D in that order. Point P is strictly inside it.
Quantity A: The combined areas of triangles PAB and PCD
Quantity B: Half the rectangle's area

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

86. Two congruent triangles have exactly one common vertex and no common entire edge. Their interiors may or may not overlap.
Quantity A: The total area they cover, counting any overlap only once
Quantity B: The sum of their individual areas

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

87. To obtain each next term, halve the current term, then use the smallest multiple of 4 that is at least that result. What are the next TWO terms? 100, 52, 28, 16, 8, ____, ____

A) 4, 0

B) 4, 4

C) 4, 2

D) 8, 4

88. The first term is 1. For term number n after that, add n to the preceding term and keep only the units digit. What are the next TWO terms? 1, 3, 6, 0, 5, 1, 8, 6, ____, ____

A) 5, 4

B) 4, 4

C) 5, 6

D) 5, 5

89. Each next term moves the first digit to the end, then adds 1. All terms shown have three digits. What comes next? 123, 232, 323, 234, 343, ____

A) 434

B) 433

C) 344

D) 334

90. Describe each consecutive group of identical digits by writing its count followed by that digit. For example, 44 becomes 24. Apply this rule repeatedly. What follows 4, 14, 1114, 3114, ____?

A) 311114

B) 131114

C) 132114

D) 3214

91. A machine first rounds its input to the nearest integer, then squares that integer. Which input produces an output of 25?

A) 4.4

B) −4.6

C) −5.6

D) 2.4

92. Two quantities have ratio 2:5. The FIRST quantity is multiplied by 3, while the second is unchanged. What is their new ratio?

A) 6:5

B) 2:15

C) 6:15

D) 5:6

93. Three prime numbers have a sum of 20. What is their greatest possible product?

A) 130

B) 144

C) 180

D) 154

94. A coin has the same probability p of heads on every flip, where 0 < p < 1. Flips are independent. H means heads and T means tails.
Quantity A: The probability of HTT
Quantity B: The probability of TTH

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

95. The real numbers x and y are both greater than 1.
Quantity A: xy + 1
Quantity B: x + y

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

96. Three positive observations have a range of 8.
Quantity A: Their mean
Quantity B: Their median

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

97. A triangle has positive area. An altitude is drawn from a vertex to the LINE containing the opposite side.
Quantity A: The altitude’s length
Quantity B: The triangle’s longest side length

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

98. A regular polygon has all its vertices on a circle of radius r, but its number of sides is not specified.
Quantity A: One side of the polygon
Quantity B: r

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

99. A paper triangle is cut into four pieces. All pieces are rearranged into a rectangle with no gaps, overlaps, stretching, or discarded material.
Quantity A: The original triangle's area
Quantity B: The rectangle's area

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

100. Start with 81. Divide by 3, then add 1, and repeat. What comes next? 81, 28, 31/3, 40/9, ____

A) 49/27

B) 67/27

C) 43/9

D) 121/27

101. Each new term is the sum of the two preceding terms MINUS 1. What comes next? 3, 5, 7, 11, 17, ____

A) 28

B) 29

C) 26

D) 27

102. Write each positive integer n followed immediately by its square, with no separator. The entries begin 11, 24, 39, 416, 525. What is the seventh entry?

A) 749

B) 736

C) 714

D) 649

103. An integer is multiplied by 6. Its result is then divided by 4, giving 15. What was the integer?

A) 6

B) 9

C) 10

D) 12

104. The digits 2, 4, and 7 are each used once to make a three-digit integer. Which choice correctly counts and lists the even integers?

A) Four: 274, 472, 724, and 742

B) Two: 274 and 724

C) Three: 247, 274, and 472

D) Six: all arrangements

105. For positive integers a and b, a/b = 5/7 and a + b = 48. What is b − a?

A) 2

B) 8

C) 12

D) 16

106. A positive integer has exactly TWO different positive factors other than 1 and itself. Which number could it be?

A) 16

B) 25

C) 30

D) 21

107. For a nonnegative integer n, consider the ODD integers strictly between 2n and 2n + 10.
Quantity A: How many such integers there are
Quantity B: 5

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

108. A positive integer n is divisible by 6.
Quantity A: n/6
Quantity B: 6

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

109. The numbers a and b satisfy 0 < a < 1 and 0 < b < 1.
Quantity A: a/(1 − b)
Quantity B: a

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

110. Two circles have radii 2 and 3 units, and their centers are 9 units apart.
Quantity A: The shortest distance between the circles
Quantity B: 5 units

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

111. A cube is divided into two pieces by a plane of symmetry.
Quantity A: The volume of one piece
Quantity B: The volume of the other piece

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

112. A circle has positive radius r.
Quantity A: The length of a quarter-circle arc
Quantity B: r

A) Quantity A is greater.

B) Quantity B is greater.

C) The two quantities are equal.

D) The relationship cannot be determined from the information given.

Section 3: Reading Comprehension

62 questions | Recommended time: 35 minutes

Literary Passage: A Very Public Stretch

At the harbor gate, Mira spotted a blue coat across the quay. Her uncle owned a coat exactly that color, and the man inside this one had raised his arm. She waved enthusiastically. He waved again, more urgently. Mira answered with both hands.

Then she noticed that his eyes were fixed above her head. A crane was lowering a cargo net behind her, and the man was directing its operator. His gestures had nothing to do with her greeting. She lowered one hand and converted the other into an elaborate stretch, as if loosening her shoulder had been her intention all along.

The man brought his palm down. The net settled. Mira, who had now stretched enough to prepare for a swimming race, inspected the timetable with extraordinary concentration. She could not decide whether she was more embarrassed by having mistaken a stranger for her uncle or by having invented a conversation entirely out of cargo signals.

Someone touched her elbow. Her uncle stood beside her, carrying his blue coat over his arm.

"I was trying to wave," he said, "but you seemed very interested in the departure times."

Mira considered explaining. Instead, she took his bag. "A fascinating subject," she said, and began walking.

113. What first leads Mira to treat the man's gesture as a greeting for her?

A) He calls her name from across the quay.

B) His coat is the same color as her uncle's, and his arm is raised.

C) He carries the bag she expects her uncle to bring.

D) Her uncle has told her to watch for a cargo signal.

114. Why does Mira turn her raised arm into a stretch after noticing the crane?

A) She hopes to help the operator lower the cargo.

B) She has suddenly decided to enter a swimming race.

C) She wants the stranger to recognize her uncle's coat.

D) She wants her mistaken wave to look like an intentional exercise.

115. In the third paragraph, inspected most nearly means:

A) rearranged in sequence

B) copied from memory

C) examined carefully

D) hid from view

116. How does the reference to preparing for a swimming race contribute to the passage?

A) It reveals the actual destination of the cargo net.

B) It establishes that Mira is training for an athletic event.

C) It humorously exaggerates the amount of stretching Mira does to cover her mistake.

D) It explains why her uncle has arrived without his coat.

117. Which description best traces Mira's changing behavior through the encounter?

A) Confident greeting, embarrassed disguise, then avoidance of a full explanation

B) Careful observation, correct identification, then a detailed report

C) Fear of the crane, relief at its departure, then anger at her uncle

D) Indifference to strangers, concern for lost cargo, then a successful rescue

Informational Passage: A Bend Without a Break

Put a straight straw into a clear glass of water and look at it from the side at an angle. The straw may seem to bend where it enters the water. Remove it, however, and it is as straight as before. The change is in the light reaching your eyes, not in the straw.

Light travels at different speeds through water and air. When it crosses the boundary at an angle, its direction changes. This bending of the light is called refraction. Light from the submerged section follows a different route to your eye from light coming from the part above the surface. Your visual system interprets the arriving light as though it had traveled in straight lines, placing the underwater section at an apparent position different from its actual one.

Viewing direction matters. Light crossing the boundary straight on does not change direction there, even though its speed changes. Thus, a change in speed does not always produce an apparent bend.

The demonstration separates an object from its appearance. Looking at the straw after removing it provides a check: the apparent bend disappears without any straightening force being applied. A report about where an object seems to be must therefore include how the observer is looking at it.

118. According to the passage, under which condition does light change direction at the water-air boundary?

A) It crosses the boundary at an angle rather than straight on.

B) It remains entirely within the straw.

C) Its speed stays unchanged in the two materials.

D) The observer first removes the straw from the glass.

119. What does the author chiefly explain through the straw demonstration?

A) A change in the path of light can alter an object's apparent shape without changing the object.

B) Water permanently changes the shape of most solid objects.

C) A visual impression is always useless as evidence.

D) Light has the same speed in every transparent material.

120. What does the example of light crossing straight on show about the explanation?

A) The speed of light cannot change unless the straw bends.

B) Every viewing direction must produce the same apparent position.

C) A speed change alone is insufficient to guarantee a change in direction.

D) The change in direction is independent of the angle of crossing.

121. In the explanation of the underwater section, apparent most nearly means:

A) physically damaged

B) completely invisible

C) deliberately invented

D) perceived rather than necessarily actual

122. What role does removing the straw play in the final paragraph?

A) It adds an unrelated technique for making drinking straws.

B) It supplies a check on the explanation by separating a lasting physical change from a viewing effect.

C) It introduces a new claim that the glass is bent.

D) It replaces the earlier explanation with a story about an observer.

Humanities Passage: A Copy With Its Own Job

Imagine an ancient coin whose worn design is difficult to see. A maker produces a clearly marked replica with sharper raised lines. Visitors can handle the replica, turn it toward a lamp, and discover a figure that is hard to recognize on the original. The copy may be easier to study in this particular way. It has not, however, become the

ancient coin.
The distinction concerns history as well as appearance. Marks on the original may record its use, the material available to its maker, or later damage. Marks deliberately added to a new replica demonstrate what the modern maker chose to reproduce. Even a very accurate copy cannot acquire the earlier object's actual journey merely by resembling it.
This does not make copying a worthless activity. A replica answers some questions while leaving others to the original. It can help explain the arrangement of a design, permit handling, or show a proposed reconstruction. A label should say which of these jobs it performs and where the reconstruction involves uncertainty.
Calling a replica a replacement is therefore like calling a rehearsal the performance that already happened. A rehearsal may reveal how an action could have been carried out; it does not become that earlier event. The useful question is not simply whether a copy is good, but what it is good for.

123. Which advantage of the imagined replica is explicitly described in the first paragraph?

A) Its metal proves which materials ancient makers always used.

B) Its label gives it the same historical journey as the original.

C) Its worn surface establishes how the original was damaged.

D) Its sharper raised lines make a difficult design easier to examine by handling and changing the light.

124. According to the author, what should a replica's label explain?

A) Only the name of the person who bought it

B) Its intended purpose and any uncertainty in the reconstruction

C) That its historical journey exactly matches the original's

D) Why visitors should never examine the original object

125. Which statement best expresses the author's view of replicas?

A) They can serve valuable purposes when their abilities and historical limits are clearly distinguished.

B) They become originals whenever their designs are sufficiently accurate.

C) They are useful only if they conceal their modern origin.

D) They should replace original objects because sharper designs contain all relevant evidence.

126. A replica exactly reproduces a scratch visible on the original coin. What would the author most strongly support?

A) Both scratches must have been caused by the same ancient event.

B) The replica has therefore acquired the original coin's age.

C) The original scratch has ceased to provide historical evidence.

D) The reproduced scratch shows a modern choice to imitate the mark, not the event that scratched the original.

127. In the third paragraph, reconstruction most nearly means:

A) the destruction of an old object

B) an attempted recreation of an earlier form

C) a record of every visitor

D) the purchase of a valuable coin

128. Why does the author compare a replica with a rehearsal in the final paragraph?

A) Both always reproduce every detail of the original without uncertainty.

B) Both are valuable only when no audience can see them.

C) Both can help someone understand an earlier thing or event without becoming that historical original.

D) Both prove that the original was created for entertainment.

Literary Passage: Where the Ink Was

"Just slide your hand across," I told Sami. "The pen will follow."
My cousin tried. A blue streak appeared below the sentence, where the side of his left hand had passed through the wet ink. I pushed a clean sheet toward him and demonstrated again with my right hand.
"Just keep your hand here."
He looked from my fingers to his own. "That is where mine was."
I was about to repeat the demonstration when he turned the paper to a different angle. He wrote his name slowly. This time his hand crossed less of the fresh writing, and the letters remained clear. He shifted the sheet a little more and tried again.
I had prepared a whole explanation about smooth strokes and graceful curves. It suddenly seemed that I had been describing what my hand already knew, without looking at what his hand was doing.
"Could you show me that angle?" I asked.
Sami turned the page back and repeated the movement. I watched the space between the nib and his wrist instead of watching my own example. Then we tested another sheet together.
On the instruction card I had made, the first line began, "Just hold the pen naturally." I crossed out Just. After a moment, I crossed out naturally too. There was room below for a more useful beginning.

129. What directly causes the blue mark below Sami's sentence?

A) The narrator deliberately draws a line under his name.

B) The paper tears along a folded edge.

C) The side of his hand passes through ink that is still wet.

D) The pen falls from the desk onto the clean sheet.

130. What change in understanding is central to the narrator's experience?

A) Graceful handwriting matters more than readable instructions.

B) A person should avoid teaching any skill that comes easily.

C) Effective guidance requires observing another person's actions rather than assuming one's own familiar movements will work for everyone.

D) Repeating the same demonstration more forcefully always solves a difficulty.

131. Why is the narrator's question "Could you show me that angle?" a meaningful turning point?

A) It proves that Sami has secretly used the pen for years.

B) It shifts responsibility for the wet ink to the paper manufacturer.

C) It shows that the narrator wants to end the lesson immediately.

D) It recognizes Sami's useful discovery and makes the narrator willing to learn from him.

132. In the phrase "a blue streak appeared," streak most nearly means:

A) a long, narrow mark

B) a period of good luck

C) an uninterrupted series of wins

D) a sudden dash across a room

133. What is the effect of ending with the narrator crossing words out of the instruction card?

A) It shows that all written directions are being permanently abandoned.

B) The visible revision gives a concrete form to the narrator's changed attitude about what a learner should find obvious.

C) It reveals that Sami had secretly written the original card.

D) It changes the story into a set of instructions for repairing a pen.

Informational Passage: Why a Full Moon Usually Stays Bright

A full Moon and a lunar eclipse are related, but they are not the same event. At full Moon, we see the Moon's sunlit side facing Earth. During a lunar eclipse, the Moon passes through Earth's shadow, so Earth blocks some or all of the direct sunlight that would otherwise reach it.

This distinction raises a question. If Earth is between the Sun and Moon near full Moon, why does an eclipse not occur every month? The Moon's orbit is tilted relative to Earth's orbital plane. At most full Moons, the Moon passes above or below the central region of Earth's shadow instead of entering it.

The missing ingredient is a sufficiently precise alignment in space. An overpass and the road beneath it can seem to cross in a flat drawing even though they occupy different heights. Similarly, a simple flat diagram of the Sun, Earth, and Moon can hide the tilt that keeps the Moon outside the shadow.

Only when the Moon's position brings it into Earth's shadow near full Moon does a lunar eclipse occur. Full Moon is therefore a necessary condition for this event, but it does not by itself guarantee the event. The explanation depends on both the illuminated side we can see and the Moon's actual path through three-dimensional space.

134. According to the passage, where does the Moon usually travel at full Moon in relation to Earth's central shadow?

A) Inside it for the entire month

B) Above or below it rather than through it

C) Behind the Sun instead of near Earth

D) Along a path that has no tilt at all

135. What is the author chiefly explaining?

A) Why Earth's shadow causes every ordinary phase of the Moon

B) Why the Sun stops shining during every full Moon

C) Why a full Moon can occur without the precise spatial alignment needed for a lunar eclipse

D) Why a flat diagram always gives more information than a spatial description

136. Suppose the Moon's orbit had no tilt and the Moon crossed Earth's shadow at every full Moon. What would follow from the passage's explanation?

A) The Moon could never appear full.

B) Lunar eclipses would occur at every full Moon under those stated conditions.

C) Earth would no longer block sunlight.

D) Lunar eclipses would be unrelated to the Moon's position.

137. How does the passage organize its explanation of full Moons and lunar eclipses?

A) It distinguishes two events, poses the apparent puzzle, and resolves it through orbital tilt and spatial alignment.

B) It lists unrelated observations and leaves their relationship unexplained.

C) It gives a historical biography followed by instructions for drawing the Moon.

D) It argues that a full Moon and an eclipse are identical, then repeats that claim.

138. What does the comparison with an overpass help readers understand?

A) Roads and the Moon must move at the same speed.

B) Apparent alignment in a flat view does not ensure that objects occupy the same level in space.

C) An eclipse can occur only directly above a bridge.

D) The height of an overpass determines the tilt of the Moon's orbit.

Informational Passage: Charges That Need Room to Move

A crystal of ordinary table salt contains electrically charged particles called ions. Yet a solid salt crystal does not conduct an electric current in the way a salt solution can. The presence of charge is only part of the explanation. A current also requires charged particles that can travel through the material.

In the solid crystal, positive and negative ions are held in an orderly structure. They can vibrate around their

positions, but they cannot move freely through the crystal. When the salt dissolves in water, the ions separate from that fixed arrangement and become mobile. Under an applied voltage, positive and negative ions move in opposite directions and carry current through the solution.
Melting salt also releases ions from the rigid solid arrangement. Thus, the important change is not the creation of charge from nothing. It is the availability of movement for charges that were already present.
Dissolving alone, however, is not enough to explain every solution. Sugar generally separates into neutral molecules when it dissolves. It does not provide the same abundant supply of mobile ions merely because it disappears from sight in water.
The examples distinguish three ideas that can easily be confused: being electrically charged, being able to move through a material, and being dissolved. To explain conduction, one must identify the particles and what they are free to do.

139. What prevents ions in the solid salt crystal from carrying current freely through it?

A) They contain no electrical charge until water touches them.

B) They have all changed into neutral sugar molecules.

C) They can move through the crystal only in one fixed direction.

D) They are held in positions within the crystal structure rather than free to travel through it.

140. What specific contrast does the passage make between dissolved salt and dissolved sugar?

A) Salt supplies mobile ions, whereas dissolved sugar generally consists of neutral molecules.

B) Salt stays visible, whereas sugar must form a solid crystal.

C) Sugar creates every charge present in the water.

D) Both substances always produce the same supply of mobile ions.

141. Which statement best captures the explanation of salt's electrical behavior?

A) Any substance that disappears in water necessarily conducts equally well.

B) Solid salt has no charged particles until they are manufactured by melting.

C) The motion of neutral molecules alone explains all electrical conduction.

D) Charge must be available in particles that can move through the material; charge being present is insufficient.

142. Why can melting salt and dissolving salt support the same central explanation even though the processes differ?

A) Both must create new charge in previously uncharged particles.

B) Both necessarily turn salt into sugar.

C) Both show that a solid structure never restricts motion.

D) Both make existing ions able to move beyond their fixed places in the solid structure.

143. In the second paragraph, mobile most nearly means:

A) containing no electrical charge

B) visible without magnification

C) able to move from place to place

D) permanently attached to one position

144. Why does the passage begin by noting that solid salt already contains charged particles?

A) It establishes an apparent puzzle that is resolved by distinguishing the presence of charge from the ability of charges to travel.

B) It proves that all later examples must involve the creation of new charge.

C) It supplies an unrelated historical detail about how salt was discovered.

D) It makes the later distinction between salt and sugar unnecessary.

Literary Passage: Between Floors

I used to believe I kept the building together. A finger touched one of my buttons, my doors closed, and I delivered people to the floor I had illuminated. Without me, surely, each floor would drift into a separate little world.

Then a maintenance worker fastened a notice to my doors: "Out of service until noon."

At first I enjoyed the importance of being unavailable. Footsteps approached, stopped, and retreated. Someone sighed. A young man with a large parcel settled down to wait nearby. This was encouraging. The building clearly needed me.

But other sounds traveled through the stairwell. A woman called a greeting upward. Two neighbors paused on a landing to compare the plants they were carrying. Someone laughed about having taken the wrong flight of stairs. The floors, to my considerable surprise, were continuing their conversations.

I began to suspect that I had confused carrying people with deciding where they belonged.

At noon the worker removed the notice. The young man stood, gathered his parcel, and pressed Five. I was pleased to be useful again. Yet the glowing number no longer looked quite like an order I had issued. It looked like a request I could answer.

I closed my doors and carried him upward. An elevator has a respectable job. Apparently, it need not also govern the building.

145. What causes the narrator to stop carrying passengers during the morning?

A) A maintenance worker places an out-of-service notice on its doors.

B) The young man orders it to remain on the ground floor.

C) Every resident moves out of the building.

D) A plant blocks every stairway.

146. What does the narrator chiefly learn during the interruption?

A) A useful service must prevent people from making requests.

B) The building has permanently lost every reason to use an elevator.

C) Being useful to others is different from controlling their choices or being the sole connection between them.

D) People choose their destinations only after seeing a lighted button.

147. Why do the conversations in the stairwell challenge the narrator's original belief?

A) They prove that every person prefers stairs to elevators.

B) They show that the maintenance worker has abandoned the building.

C) They establish that the waiting parcel contains plants.

D) They show that people maintain connections between floors even while the elevator is unavailable.

148. Which feature most clearly establishes personification in the passage?

A) The story includes a notice with a time on it.

B) The passenger has a parcel to carry.

C) The elevator describes feeling important, being surprised, and revising its own beliefs.

D) The building has a stairwell and several floors.

149. Which description best traces the elevator narrator's changing outlook?

A) Boredom becomes anger, then a determination to stop working forever.

B) Uncertainty becomes proof that the narrator controls every resident.

C) Fear becomes a plan to replace all the building's stairs.

D) Self-importance gives way to surprise, then to a more modest satisfaction in useful work.

Literary Passage: A Longer View

Niko missed the mountains most when he looked for the weather. At home, he studied a notch between two peaks before deciding whether to carry a coat. During his visit to Grandmother on the plains, there was no familiar gap to watch. From her window he could see grass, a fence, and more grass.

"You look like someone waiting for a wall," Grandmother said.

"At least a wall would be something to look at."

She did not argue. She pulled a chair beside the window and began sorting a basket of beans.

Later that afternoon, a dark band formed far beyond the fence. Beneath it hung a gray curtain, narrow at first and then wider. Niko leaned toward the glass.

"It's raining over there."

"Yes."

"But not here."

"Not yet."

He watched the distant rain move across the open country. At home, the mountains would have hidden that part of the sky. Here he could follow the approaching change for a long time before a single drop touched Grandmother's window.

The grass bent in a broad line. The first drops finally arrived, and Grandmother moved her basket farther from the sill.

That night, Niko started to draw the curtains, then stopped. He left a wide space between them. The fence was almost invisible in the dusk, but the distance beyond it no longer seemed empty.

150. What does Niko first notice beneath the distant dark band?

A) Snow collecting in the notch between nearby peaks

B) A gray curtain of rain beyond the fence

C) Grandmother walking across a distant bridge

D) A wall being built around the house

151. Which statement best captures Niko’s change during the visit?

A) He decides that weather never changes on the plains.

B) He discovers that his memory of the mountains was entirely false.

C) He concludes that Grandmother must move to the mountains.

D) He begins to value a feature of the unfamiliar landscape that his first comparison with home had made him overlook.

152. Why is Niko’s decision to leave a space between the curtains significant?

A) He is certain the curtains will stop the rain.

B) He wants to conceal the fence from Grandmother.

C) He now expects the open view to offer something worth noticing.

D) He has forgotten that the house has windows.

153. In the first paragraph, notch most nearly means:

A) a written forecast

B) a narrow gap

C) a sudden gust

D) a marked improvement

154. What does describing the distant rain as a gray curtain help readers picture?

A) A fabric barrier Grandmother has put outside the window

B) A hanging, sheetlike area of rain that gradually widens across Niko’s view

C) A mountain that disappears permanently

D) A narrow beam of sunlight crossing the floor

Informational Passage: Two Questions About a Wing

Animal body parts can be grouped in more than one useful way. If the question is which parts assist flight, a bird's wing and a butterfly's wing belong together. That group answers a question about function. It does not, by itself, show that the wings have the same internal construction.

A bird's wing contains upper-arm and forearm bones that correspond to bones in a human arm. Their proportions and uses differ, but their positions and connections reveal a shared structural pattern. A butterfly's wing does not contain that same forelimb arrangement. Thus, parts with different jobs can share a basic pattern, while parts with a similar job can be built differently.

Corresponding structures need not look identical. A bone may be lengthened, shortened, or differently shaped while retaining its place in an arrangement. Looking only at outward shape or at what a part does can therefore hide important anatomical relationships.

Neither grouping rule is useless. Someone studying the different ways animals fly may properly begin with function. Someone comparing underlying anatomy must ask about structures and connections instead. Claims about evolutionary relationships also require evidence beyond a single shared job. Before sorting the examples, a reader should identify the question that the proposed group is supposed to answer.

155. Which shared structural feature does the passage identify in a bird’s wing and a human arm?

A) Identical outward shapes and proportions

B) Corresponding upper-arm and forearm bones

C) The same ability to support flight

D) The complete absence of internal bones

156. What is the main point of the passage’s comparison of body parts?

A) Every body part used for flight must have identical internal bones.

B) Only outward appearance can reveal useful similarities.

C) Grouping parts by their function can never serve a legitimate purpose.

D) A grouping criterion should fit the question being asked; similar function and similar underlying structure are different criteria.

157. An unfamiliar animal part is used for swimming but has corresponding upper-arm and forearm bones. Which conclusion best follows from the passage?

A) Its different job would not prevent a structural comparison with a human arm or bird’s wing.

B) Its swimming function proves that it has no anatomical relationships with arms.

C) It must be identical in shape to every bird’s wing.

D) Its function alone settles every question about its evolutionary relationships.

158. In the discussion of bones, corresponding most nearly means:

A) matching in relative position and relationship

B) identical in every visible detail

C) used for exactly the same job

D) unrelated despite a shared color

159. Why does the author acknowledge that grouping by function can be useful before discussing its limits?

A) To show that the first criterion answers a real question even though it cannot answer every anatomical or evolutionary question

B) To deny the later comparison between bones

C) To argue that all grouping choices give the same information

D) To make the butterfly example irrelevant to the discussion

Literary Passage: The Appointment at Four

When the market clock showed four, Sera stood beneath it and began composing a speech. Ben had promised to meet her there. She had brought the heavy box of photographs he wanted, and now she was the one left waiting.

At first the speech was polite. Then it acquired sharp remarks about promises. She bristled at each passerby who seemed to be Ben and was not.

A worker approached the clock with a narrow ladder.

"Keeping you company?" he asked.

"Keeping track of how late someone is."

"Not very reliably, then." He pointed at the hands. "We moved them forward while testing the mechanism. I still have to put them back. It is not quite four yet."

Sera looked down at the box. Her prepared remarks suddenly seemed heavier than the photographs.

Ben appeared at the entrance carrying two paper cups.

"I thought you might be early," he said. "Would you like tea?"

Sera nearly said that she had only just arrived. It was an attractive answer: short, harmless, and impossible to check. Instead, she set the box down.

"I was early," she said. "And I was getting ready to blame you for it."

Ben glanced at the worker, then at the clock.

"Will the photographs forgive me?"

She laughed and accepted the cup. Her speech was never delivered. For once, she was pleased to have prepared something she did not use.

160. What reveals that the clock cannot establish Ben’s lateness?

A) Ben says every market clock is slow.

B) A photograph shows the correct time.

C) The worker says its hands were advanced during testing.

D) Sera finds a broken watch.

161. Why does Sera nearly claim she has only just arrived?

A) It would conceal the mistaken accusation she had been preparing.

B) She wants to hide the photographs.

C) Ben has refused to meet her.

D) The worker tells her to lie.

162. In the description of Sera’s reaction to passersby, bristled most nearly means:

A) reacted with irritation

B) sought shelter

C) counted silently

D) grew taller

163. What does making the prepared remarks seem heavier than the photographs suggest?

A) She has printed the speech inside the box.

B) The photographs have disappeared.

C) Ben has asked for a lighter box.

D) Her intended accusation has become an uncomfortable burden.

164. Which sequence best summarizes Sera's response?

A) She misses the meeting and hides the photographs.

B) She proves Ben is late and delivers her speech.

C) She prepares blame, discovers a faulty assumption, and chooses an honest admission.

D) She blames the worker and refuses the tea.

Informational Passage: A Shadow's Soft Edge

A shadow need not have one sharp boundary. With a small, distant light source, its edge may look quite definite. With a larger source nearby, a broad gray region can appear between the darkest shadow and the fully lit surface.

To understand the difference, imagine several small lights spread across the face of one large lamp. Each casts a slightly displaced shadow of the same object. At some places, the object blocks light from every part of the source. These places form the darkest region. Elsewhere, the object blocks only some parts, while light from the remaining parts still reaches the surface.

The partly shaded region is called the penumbra. Moving across it changes how much of the source is visible, so the transition in brightness can be gradual rather than abrupt.

The explanation does not require the object itself to have a fuzzy edge. A sharply cut card can cast a soft-edged shadow when the lighting geometry creates a wide region of partial blockage. Changing the size or position of the source can change that region without changing the card.

The useful question is therefore not simply whether a surface is in shadow. One must also ask how much of the source is blocked there. A shadow records relationships among the source, object, and receiving surface.

165. What distinguishes the darkest region from the penumbra in this model?

A) Only the darkest region contains an object.

B) A different lamp lights the penumbra.

C) All source light is blocked in the darkest region; only some is blocked in the penumbra.

D) The darkest region requires a fuzzy object.

166. What chiefly explains a soft shadow edge?

A) Every shadow proves its object has an uneven edge.

B) Large lamps cannot produce shadows.

C) Every shadow point receives equal light.

D) Partial blockage of an extended source creates a gradual brightness transition.

167. A sharply cut card produces a broad gray shadow boundary. What is best supported?

A) The card must have changed shape.

B) The surface receives no light anywhere.

C) The shadow alone does not establish that the card's edge is physically fuzzy.

D) The source must be a single tiny point.

168. The passage uses penumbra to name:

A) a lamp casing

B) a region receiving light from only part of the source

C) a completely dark object

D) the brightest point on a lamp

169. Why does the author imagine separate small lights across one lamp?

A) To give lamp assembly instructions

B) To prove that all lamps contain separate bulbs

C) To replace shadow observations with a lamp count

D) To distinguish contributions from different positions and explain partial blockage

Humanities Passage: The Door at the End

Suppose a story follows a traveler returning to a house after a long disagreement with its occupants. Earlier, the traveler insists that carrying a key is enough: there will be no need to ask anyone for admission. In the final scene, the traveler reaches the door, puts the key away, and knocks. The story ends before anyone answers.

A reader may object that the ending leaves the most obvious question unresolved. Who opens the door, and what will happen next? Those questions remain open. Yet the story has answered another question: how the traveler now approaches the people inside. The gesture of knocking differs meaningfully from the earlier insistence on using the key.

Such an ending can withhold the outcome of an encounter while completing a change in attitude. Its lack of a final explanation need not mean that nothing has been achieved. The reader must identify which uncertainty remains and which development the last action makes visible.

Nor does an open ending make every interpretation equally persuasive. Predicting that the house will turn into a ship requires support this example has not supplied. An interpretation about changed expectations can point to the contrast between the key and the knock. An unfinished event still belongs to a shaped work, and a reader's freedom remains connected to its evidence.

170. What distinction is central to the discussion?

A) Every unanswered question makes a story meaningless.

B) An action has no significance without dialogue.

C) All predictions about an open ending are equally supported.

D) An encounter can remain unresolved while a character's changed attitude becomes clear.

171. Why is an interpretation about changed expectations stronger than the ship prediction?

A) It invents more events.

B) It connects the final gesture with the traveler's earlier stated attitude.

C) It depends on knowing the traveler's name.

D) It ignores the key and considers only the house's size.

172. In the third paragraph, withhold most nearly means:

A) leave undisclosed

B) make stronger

C) repeat exactly

D) prove impossible

173. How do the key and the knock function in the essay?

A) They show the traveler has forgotten how doors work.

B) Their contrast makes a change in the traveler's attitude visible through action.

C) They identify who answers the door.

D) They only establish the building's size.

174. Which question does the example expressly leave unanswered?

A) Whether the traveler reaches the house

B) Who answers the knock and what happens afterward

C) Whether the traveler possesses a key

D) Whether the traveler knocks

Section 4: Mathematics

64 questions | Recommended time: 45 minutes

175. Define the fractional part of a number as the number minus the greatest integer less than or equal to it. What is the fractional part of −2.7?

A) 0.3

B) −0.7

C) 0.7

D) −0.3

176. A marker begins at −4 on a number line, moves 7 units right, then moves left by HALF the distance it just moved right. Where does it finish?

A) −3.5

B) −0.5

C) 0.5

D) 3.5

177. Digits a and b form the decimals a.b and b.a, where the digit after the point is in the tenths place. If a.b + b.a = 12.1, what is a + b?

A) 10

B) 11

C) 12

D) 13

178. Two-thirds of a batch of tiles are marked. Five marked tiles break and are removed; no other tiles are removed. Then three-fifths of the remaining tiles are marked. How many tiles were in the original batch?

A) 30

B) 15

C) 25

D) 45

179. Define f(x) = 2x + 1 and g(x) = x + k. What value of k makes f(g(x)) = g(f(x)) for every real x?

A) 1

B) −2

C) 0

D) 3

180. For which values of x is (x − 1)/(x + 2) negative?

A) x < −2

B) x > 1

C) All x < 1

D) −2 < x < 1

181. Someone claims that $n^2 + n + 1$ is prime for every positive integer n. Which pair gives the SMALLEST counterexample n and the resulting composite value?

A) n = 1; value = 3

B) n = 2; value = 7

C) n = 3; value = 13

D) n = 4; value = 21

182. Which set contains ALL real solutions of $|x - 2| + |x - 7| = 5$?

A) Only x = 2 and x = 7

B) Every x from 2 through 7, including both endpoints

C) Every x greater than or equal to 7

D) No real values of x

183. A rectangle is 3 units wide and 5 units long. Which statement about reflection across a diagonal is true?

A) The diagonal is a line of symmetry for every rectangle.

B) Every point on the diagonal moves to a different point.

C) Reflection changes the rectangle's area from 15 to 25 square units.

D) The reflected rectangle does not coincide with the original because this rectangle is not a square.

184. A right triangle has legs 6 and 8 centimeters. A semicircle is built outward on its hypotenuse, using the hypotenuse as diameter, with no overlap of interiors. What is their combined area?

A) 24 + 50π square centimeters

B) 24 + 25π/2 square centimeters

C) 48 + 25π/2 square centimeters

D) 24 + 25π square centimeters

185. A cube rests on a flat surface. Its faces initially point up, down, north, south, east, and west. It rolls one quarter-turn east, one quarter-turn north, then TWO quarter-turns east. Each roll is over an edge, without twisting. Which ORIGINAL face is now on top?

A) The face originally pointing south

B) The face originally pointing east

C) The face originally pointing north

D) The face originally pointing down

186. A moving walkway is 100 meters long. A traveler walks at 1 meter per second relative to its belt. For the first 20 seconds, the belt moves forward at 0.5 meter per second; it then stops. The traveler keeps walking at the same speed relative to the belt. How long does the trip take?

A) 80 seconds

B) 100 seconds

C) 120 seconds

D) 90 seconds

187. A 5-minute hourglass and an 8-minute hourglass begin running together. At minute 5, the empty 5-minute glass is flipped. At minute 8, the empty 8-minute glass is flipped. When the 5-minute glass next empties, the 8-minute glass is immediately flipped again. At what time after the original start will the 8-minute glass now empty?

A) 12 minutes

B) 10 minutes

C) 13 minutes

D) 16 minutes

188. A tile numbered from 1 through 8 is chosen uniformly at random. Independently, a spinner selects red or blue. Given that the spinner shows blue, what is the probability that the tile's number is divisible by 4?

A) 1/4

B) 1/8

C) 1/2

D) 1/3

189. Fourteen repair times, measured in whole minutes, are grouped as follows: 0–9 minutes, 4 repairs; 10–19 minutes, 7 repairs; 20–29 minutes, 3 repairs. What must be true about the median?

A) It is exactly 14 minutes.

B) It is less than 10 minutes.

C) It is between 10 and 19 minutes, inclusive.

D) It is greater than 19 minutes.

190. Four separate tiles bear the letters O, B, O, and E. They are thoroughly mixed, then placed in a row by uniformly random ordering of the four physical tiles. What is the probability that the visible row spells OBOE?

A) 1/24

B) 1/6

C) 1/12

D) 1/4

191. A red token has value −2 and a blue token has value +3. Which collection has a total value of zero?

A) 2 red tokens and 3 blue tokens

B) 1 red token and 1 blue token

C) 1 red token and 2 blue tokens

D) 3 red tokens and 2 blue tokens

192. The greatest common factor of 18 and a positive integer b is 6. Their least common multiple is 90. What is b?

A) 30

B) 15

C) 45

D) 60

193. How many positive integers n from 1 through 50 make √(18n) an integer?

A) 3

B) 7

C) 10

D) 5

194. What value of k makes the equations 3x + 6 = 12 and kx + 4 = 10 have the SAME solution for x?

A) 6

B) −3

C) 3

D) 12

195. A sorting tray has n rows, with n + 4 wells in every row, where n > 2. The first and last rows are reserved; all remaining wells are used. Which expression gives the number of used wells?

A) $n^2 + 4n - 2$

B) $n^2 - 8$

C) $n^2 + 6n + 8$

D) $n^2 + 2n - 8$

196. The formula A = 2h(w + d) relates four quantities, with h ≠ 0. Which expression equals w?

A) A/(2h) − d

B) A/(2h − d)

C) A/(2h) + d

D) (A − d)/(2h)

197. The positive integer x satisfies BOTH x + 4 > 2x − 3 and $x^2 > 25$. Which set contains all possible values of x?

A) x = 5 only

B) x = 7 only

C) x = 6 only

D) x = 5 and x = 6

198. Two square-based pyramids are joined base to base. The shared square bases are entirely inside the resulting solid. Which description gives all the exposed faces?

A) 8 triangular faces and 2 square faces

B) 4 triangular faces and 2 square faces

C) 8 triangular faces

D) 6 square faces

199. A circle of area 36π square units lies in the region x ≥ 0, y ≥ 0 and is tangent to BOTH coordinate axes. What are its center's coordinates?

A) (36, 36)

B) (6, 6)

C) (6, 0)

D) (0, 6)

200. A cube has edge length 4 units. A sphere has the same center as the cube, and all eight cube vertices lie on the sphere. What is the sphere's radius?

A) $4\sqrt{3}$ units

B) $2\sqrt{3}$ units

C) $2\sqrt{2}$ units

D) 4 units

201. A parcel may weigh at most 2 pounds. Its two parts weigh 13 ounces and 1 pound 1 ounce; wrapping adds 3 ounces. Does the wrapped parcel meet the limit? Use 16 ounces = 1 pound.

A) Yes; it weighs 30 ounces.

B) Yes; it weighs exactly 2 pounds.

C) No; it exceeds the limit by 3 ounces.

D) No; it exceeds the limit by 1 ounce.

202. For this problem, charge used in milliampere-hours equals current in milliamperes multiplied by time in hours. A device uses 300 milliamperes for 40 minutes, then 120 milliamperes for 30 minutes. What total charge does it use?

A) 260 milliampere-hours

B) 420 milliampere-hours

C) 200 milliampere-hours

D) 360 milliampere-hours

203. Only the two broad faces of a 25-centimeter-by-40-centimeter plate receive a uniform coating 0.02 millimeter thick. Each cubic centimeter of coating weighs 1.2 grams. How much mass is added? Use 10 millimeters = 1 centimeter.

A) 2.4 grams

B) 4.8 grams

C) 48 grams

D) 0.48 gram

204. A counter starts at 0 and shows cumulative totals of 18, 47, and 65 pulses at the ends of minutes 1, 2, and 3. During which minute were the MOST new pulses recorded?

A) Minute 1

B) Minute 3

C) Minute 2

D) Each minute recorded the same number.

205. Two independent signal units each show green or red. The first shows green with probability 3/4, and the second with probability 2/3. What is the probability that they show DIFFERENT colors?

A) 5/12

B) 1/2

C) 11/12

D) 1/12

206. Machine P makes 9 acceptable parts out of 10 in the morning and 36 out of 60 in the evening. Machine Q makes 48 acceptable parts out of 60 in the morning and 5 out of 10 in the evening. Which comparison is correct?

A) P has the higher rate in each session and overall.

B) P has the higher acceptable-part rate in each session, but Q has the higher rate overall.

C) Q has the higher rate in each session and overall.

D) Their overall rates are equal because both make 70 parts.

207. Which decimal writes fifteen hundredths with an extra final zero that does NOT change its value?

A) 1.50

B) 0.0150

C) 0.150

D) 15.0

208. A dial moves from zero only in steps of 1/8 unit. Which setting is NOT reachable?

A) 0.70

B) 0.125

C) 0.375

D) 0.625

209. An indicator shows 6 milliliters at the one-quarter mark of a container. What fraction of the full capacity is 4 milliliters?

A) 1/4

B) 2/3

C) 1/24

D) 1/6

210. What is the greatest positive two-digit integer whose square ends in 44?

A) 92

B) 98

C) 82

D) 88

211. A function satisfies F(a + b) = F(a) + F(b) + 1 for every pair of inputs. If F(2) = 5, what is F(6)?

A) 15

B) 17

C) 11

D) 18

212. The expressions a + b and a − b have values 9 and 4, respectively. What is $a^2 - b^2$?

A) 5

B) 13

C) 36

D) 65

213. For which real values of x is $\sqrt{(x-1)/(x-4)}$ defined as a real number?

A) $x > 1$

B) $x \geq 1$ and $x \neq 4$

C) $x \geq 4$

D) Every real x except 4

214. Consider all four equations together: a + b = 8; b + c = 11; c + d = 9; d + a = 7. Which statement is true?

A) A unique set of real values satisfies them.

B) Infinitely many sets of real values satisfy them.

C) They are true for every choice of a, b, c, and d.

D) No set of real values satisfies all four equations.

215. Point P is (−3, 5). Point Q lies on the y-axis at the SAME height as P. What are Q's coordinates?

A) (−3, 0)

B) (0, −3)

C) (5, 0)

D) (0, 5)

216. One transformation moves a point 2 units right. A different transformation reflects it across the vertical line x = 2. Which point lands at the SAME location under either transformation?

A) (0, 4)

B) (1, 4)

C) (2, 4)

D) (3, 4)

217. A rectangle has long side L and short side W. It is cut into three congruent rectangles, each with dimensions L/3 and W. Each small rectangle has long side W and is similar to the original. What is L/W?

A) 3

B) 3/2

C) $\sqrt{3}$

D) $\sqrt{2}$

218. Clock B is always 2 hours ahead of Clock A. A process starts when Clock A reads 11:50 a.m. and lasts 25 minutes. What does Clock B read when the process ends?

A) 2:15 p.m.

B) 12:15 p.m.

C) 1:15 p.m.

D) 2:25 p.m.

219. Ruler A has marks every 1/20 centimeter. Ruler B has marks every 0.4 millimeter. Which ruler has the finer spacing, and by how much? Use 1 centimeter = 10 millimeters.

A) Ruler B, by 0.1 millimeter

B) Ruler A, by 0.1 millimeter

C) Ruler A, by 0.35 millimeter

D) Ruler B, by 0.35 millimeter

220. An instrument shows either A or B on each of four independent trials, with each label equally likely. Compare each result after the first with the result immediately before it. What is the probability that the label CHANGES exactly twice?

A) 3/8

B) 3/16

C) 1/2

D) 3/4

221. A data set has median 8 and range 6. Every value x is replaced by 20 − 2x. What are the new median and range?

A) Median 4; range −12

B) Median 4; range 12

C) Median 16; range 12

D) Median 4; range 8

222. A fair six-sided number cube is rolled. If the result is 1, 2, or 3, that number is recorded. If the result is 4, 5, or 6, the number 1 is recorded. What is the probability that the RECORDED number is 1?

A) 1/3

B) 1/6

C) 2/3

D) 5/6

223. Which label represents a DIFFERENT value from the other three?

A) 3/5

B) 0.60

C) 60%

D) 0.06

224. For how many integers n from 1 through 83 does n/84 reduce to a fraction with denominator 7?

A) 7

B) 12

C) 6

D) 5

225. Both decimals repeat their two-digit block forever. What is 0.363636... ÷ 0.121212...?

A) 3, exactly

B) Approximately 3, but not exactly

C) 1/3, exactly

D) There is not enough information.

226. For which x is the proposed shortcut $(x + 2)^2 = x^2 + 4$ actually true?

A) x = 2 only

B) x = 0 only

C) Every real x

D) No real x

227. Given x/3 + y/2 = 7, what is 2x + 3y?

A) 42

B) 14

C) 21

D) 7

228. F(x) is the GREATER of x and 5. Which x satisfies F(x) = x + 2?

A) x = 5

B) x = 3

C) x = 7

D) Every x ≥ 5

229. For x ≠ −2 and x ≠ −1, consider x/(x + 2) = (x − 1)/(x + 1). Which statement is correct?

A) x = 0 satisfies it.

B) No real x in the stated domain satisfies it.

C) x = 1 satisfies it.

D) Every x in the stated domain satisfies it.

230. Move from (0, 0) to (3, 4) using only horizontal and vertical unit steps. What is the minimum total distance?

A) 5 units

B) 12 units

C) 7 units

D) 1 unit

231. An isosceles trapezoid has parallel bases 8 and 14 units long and equal legs 5 units long. What is its height?

A) 3 units

B) 4 units

C) 5 units

D) √34 units

232. A 4-by-4-by-4 solid consists of unit cubes. Only the TOP and FRONT faces are painted. How many unit cubes have EXACTLY ONE painted face?

A) 28

B) 32

C) 16

D) 24

233. Which 24-hour clock reading represents 9:05 p.m.?

A) 09:05

B) 20:05

C) 21:50

D) 21:05

234. A sound pulse travels to a wall and back in 0.8 second at 340 meters per second throughout. How far away is the wall?

A) 136 meters

B) 272 meters

C) 425 meters

D) 68 meters

235. A bar is 2.112 meters long. It is cut into seven equal pieces. Ends need no trimming; each separating cut removes 2 millimeters. What is each piece's EXACT length?

A) 30 6/35 centimeters

B) 29 34/35 centimeters

C) 30 centimeters

D) 300 centimeters

236. A system flags 6 of 20 packages. Four of those six actually have incorrect labels. What fraction of the FLAGGED packages are false alarms?

A) 1/10

B) 1/5

C) 2/3

D) 1/3

237. Four measurements have mean 12. A fifth measurement between 4 and 16 inclusive is added. Which NEW mean CANNOT occur?

A) 11

B) 12

C) 10

D) 12.8

238. Three fair coins are tossed independently. Given that at least one shows heads, what is the probability that EXACTLY TWO show heads?

A) 3/7

B) 3/8

C) 1/2

D) 1/3

Section 5: Language

60 questions | Recommended time: 30 minutes

239. Which sentence correctly uses commas to set off the explicit contrast not fragile?

A) The shell is thin not, fragile and it bends slightly.

B) The shell is thin, not fragile and, it bends slightly.

C) The shell, is thin not fragile, and it bends slightly.

D) The shell is thin, not fragile, and it bends slightly.

240. Which version correctly places the semicolon when joining two complete clauses, with only the word abrupt quoted?

A) The editor called the ending "abrupt;" the writer agreed.

B) The editor called the ending; "abrupt" the writer agreed.

C) The editor called the ending "abrupt" the writer; agreed.

D) The editor called the ending "abrupt"; the writer agreed.

241. Which version correctly uses a colon to introduce three listed items and a semicolon to join the two independent clauses?

A) The kit holds three essentials; a compass, a map, and a lamp, a separate pouch holds batteries.

B) The kit holds three essentials: a compass, a map, and a lamp, a separate pouch holds batteries.

C) The kit holds three essentials: a compass, a map, and a lamp; a separate pouch holds batteries.

D) The kit holds: three essentials, a compass, a map, and a lamp; a separate pouch holds batteries.

242. Which sentence correctly distinguishes a nationality adjective from the ordinary name of a Greek letter?

A) The Greek letter delta appears beside the diagram.

B) The greek letter delta appears beside the diagram.

C) The Greek Letter Delta appears beside the diagram.

D) The greek letter Delta appears beside the diagram.

243. A heading must use SENTENCE CASE: capitalize its first word and proper nouns only. Which heading follows that instruction?

A) Crossing the Atlantic in winter

B) Crossing The Atlantic In Winter

C) crossing the Atlantic in winter

D) Crossing the atlantic in Winter

244. Choose the correctly spelled name of the frozen fruit dessert.

A) sherbet

B) sherbitt

C) sherbett

D) sherbit

245. Which spelling correctly completes the sentence? The scarf was a vivid ___, a purplish-red color.

A) fuschia

B) fuchsia

C) fushcia

D) fuchisa

246. Choose the grammatically correct completion: "The steep path is worth ___."

A) to climb

B) climbed

C) climbing

D) climb

247. A flight began at noon and landed at 3 p.m. In a past-tense account, which completion uses the PAST PERFECT PROGRESSIVE to express its duration up to landing? "When we landed, we ___ for three hours."

A) have been flying

B) will be flying

C) are flying

D) had been flying

248. Which revision correctly places NOT ONLY at the beginning of the sentence? "The guide found the path and also explained how it had formed."

A) Not only the guide found the path, but the guide also explained how it had formed.

B) Not only did the guide find the path, but the guide also explained how it had formed.

C) Not only did the guide found the path, but the guide also explained how it had formed.

D) Not only does the guide found the path, but the guide also explained how it had formed.

249. Choose the pair that means CONVENIENT FOR AN IMMEDIATE PURPOSE in the first blank and QUICK AND EFFICIENT in the second: "A temporary patch was ___; the repair crew's response was ___."

A) expeditious; expedient

B) expedient; expedient

C) expedient; expeditious

D) expeditious; expeditious

250. Choose the pair that correctly completes the sentence: "The supervisor ___ authority to the deputy but ___ obsolete paperwork to storage."

A) relegated; delegated

B) delegated; relegated

C) delegated; delegated

D) relegated; relegated

Editing Passage: The Envelope That Explains Its Shape

(1) A maker sells paper ornaments in envelopes. (2) Some buyers press the envelopes flat when placing them in a bag. (3) Therefore, the maker folds a shallow ridge into a cardboard insert. (4) The ridge holds the envelope away from the ornament's delicate raised center. (5) The insert uses cardboard. (6) The package should also say, "Keep the raised side facing up; do not flatten." (7) The ridge provides physical protection, while the instruction explains how to preserve it. (8) Together, the shape and the message help the buyer carry the ornament without crushing the feature the package was designed to protect.

251. Which revision of sentence 5 adds useful information about the insert rather than merely repeating what sentence 3 already says?

A) The cardboard insert is an insert made of cardboard.

B) Many different objects are made from many different materials.

C) The maker once bought a blue pen for an unrelated drawing.

D) A small arrow printed beside the ridge can show which side should face up.

252. Which transition could replace Therefore at the start of sentence 3 while preserving its logical connection to sentence 2?

A) For a contrary reason,

B) Despite that protection,

C) To prevent this,

D) Long before this happened,

253. Which added sentence would most weaken the unity of this passage?

A) The maker prefers painting large outdoor walls to decorating small paper objects.

B) The ridge should remain higher than the raised center of the ornament.

C) Printing the instruction near the opening makes it visible before handling.

D) An arrow can match the words raised side facing up.

254. Which version correctly writes the familiar expression for an exact hour?

A) The bell rang at five oclock.

B) The bell rang at five oc'lock.

C) The bell rang at five ocloc'k.

D) The bell rang at five o'clock.

255. The quoted words are a STATEMENT, but the whole sentence is a QUESTION about that statement. Which version places the question mark correctly?

A) Did the guide write, "The gate is closed"?

B) Did the guide write, "The gate is closed?"

C) Did the guide write, "The gate is closed."?

D) Did the guide write, "The gate is closed?"?

256. Which sentence correctly distinguishes a person's name from the ordinary name of a measurement unit?

A) The force is measured in newtons, a unit named after Newton.

B) The force is measured in Newtons, a unit named after newton.

C) The force is measured in Newtons, a Unit named after Newton.

D) The force is measured in newtons, a unit named after newton.

257. Which sentence correctly capitalizes a named storm and a later generic reference to it?

A) hurricane Iris weakened overnight; the Hurricane continued north.

B) Hurricane iris weakened overnight; the hurricane continued north.

C) Hurricane Iris weakened overnight; the hurricane continued north.

D) Hurricane Iris weakened overnight; the Hurricane continued North.

258. Which spelling correctly names the pasta in the sentence? "The cook stirred the ___ into the sauce."

A) macaroni

B) macaronee

C) macaronni

D) macarony

259. Which spelling correctly names a woven decorative wall hanging?

A) tapestry

B) tapistry

C) tapestrie

D) tappestry

260. Which spelling correctly names an intermediate floor between two main floors of a building?

A) mezanine

B) mezzannine

C) mezzanene

D) mezzanine

261. The family surname is Rowley. Which plural names the family members without showing possession? "The ___ arrived together."

A) Rowlies

B) Rowleys

C) Rowley's

D) Rowleys'

262. Which form correctly completes the sentence? "The ceramics ___ by the potter on Monday are ready for firing."

A) shaping

B) were shaped

C) shaped

D) shape

263. Choose the compound pronoun required in formal written English: "___ the captain selects will guide the group."

A) Whoever

B) Whosever

C) Whomever

D) Whoever's

264. Choose the words that correctly complete the sentence: "The hikers found ___ shelter after the steep ___; there was no ___ about staying there."

A) decent; dissent; descent

B) decent; descent; dissent

C) descent; decent; dissent

D) dissent; descent; decent

265. Choose the pair that correctly completes the sentence: "The current has ___ the boat past the village where its builder was ___."

A) born; borne

B) borne; born

C) borne; borne

D) born; born

Editing Passage: Keeping a Correction Visible

(1) At an equipment desk, a log records every item issued and returned. (2) Erasing an incorrect entry removes both the error and evidence that the record changed. (3) Only after checking that the old entry is readable should the writer add the corrected information, date, and initials beside it. (4) First, draw a single line through the incorrect entry. (5) Check that the old entry can still be read through this line. (6) The corrected information shows what changed. (7) The initials show who made the change. (8) The log can then be reviewed without guessing which words used to be on the page.

266. In which order should sentences 3, 4, and 5 appear so that the stated sequence of actions is followed?

A) 3, 4, 5

B) 5, 3, 4

C) 4, 3, 5

D) 4, 5, 3

267. Which sentence best concludes the passage while preserving its purpose?

A) The log should be decorated so that it looks more expensive.

B) Because errors occur, equipment should never be issued.

C) Every incorrect entry must be erased completely before review.

D) Readable corrections let later readers check the log's history without losing the original record.

268. Which sentence best combines sentences 6 and 7 while retaining their distinct information?

A) The initials show what changed, and the corrected information shows who made the change.

B) The corrected information shows what changed, and the initials show who made the change.

C) The corrected information and the initials show exactly the same thing.

D) The initials show who made the change because mistakes never occur.

269. Which ending most clearly shows an ABRUPT INTERRUPTION of the unfinished spoken sentence, rather than a voice trailing away?

A) "I thought the answer was—"

B) "I thought the answer was..."

C) "I thought the answer was:"

D) "I thought the answer was,"

270. Two sisters-in-law jointly own a car. Which phrase correctly makes the PLURAL compound noun possessive?

A) the sisters-in-law’s car

B) the sister’s-in-law car

C) the sisters’-in-law car

D) the sisters-in-laws’ car

271. An editor replaces She with Nora in the quotation "She left at dawn." Square brackets must identify the replacement; all other quoted words stay unchanged. Which version follows that rule?

A) "[Nora] left at dawn."

B) "Nora left at dawn."

C) "(Nora) left at dawn."

D) "Nora [left at dawn]."

272. Which sentence correctly capitalizes the first-person pronoun while preserving the specified LOWERCASE letter being discussed?

A) i wrote a lowercase i below the line.

B) I wrote a lowercase I below the line.

C) I wrote a lowercase i below the line.

D) i wrote a Lowercase i below the line.

273. Which sentence correctly capitalizes the name of a specific historical period while leaving ordinary descriptive words lowercase?

A) The chapter compares bronze age metalwork with modern casting.

B) The chapter compares Bronze age metalwork with Modern Casting.

C) The chapter compares bronze Age metalwork with modern casting.

D) The chapter compares Bronze Age metalwork with modern casting.

274. Which spelling correctly names the animal with a ringed tail and dark markings around its eyes?

A) raccon

B) raccoonn

C) raccoan

D) raccoon

275. Which spelling correctly names a soft leather shoe?

A) moccasinn

B) moccasan

C) moccasim

D) moccasin

276. What is the understood subject of the complete command "Pass the towel to Ken"?

A) the towel

B) Ken

C) you

D) There is no understood subject.

277. Choose the pair that correctly completes the sentence: "The criteria ___ clear, but one criterion ___ more weight than the others."

A) is; carries

B) are; carry

C) are; carries

D) is; carry

278. Which choice is a complete, grammatically correct sentence in formal written English?

A) The final note had faded, the audience remained still.

B) The final note having faded while the audience remained still.

C) The final note having faded, the audience remained still.

D) The final note having faded, the audience remaining still.

279. Choose the pair that correctly completes the sentence: "The cook sliced a ___ while the plumber repaired a ___."

A) leak; leek

B) leek; leak

C) leek; leek

D) leak; leak

280. Choose the pair that correctly completes the sentence: "The climber became ___ after passing the enormous ___."

A) boulder; bolder

B) bolder; boulder

C) bolder; bolder

D) boulder; boulder

281. Choose the pair that correctly completes the sentence: "The reviewers ___ about the design, so they will ___ the decision until Friday."

A) defer; differ

B) differ; defer

C) differ; differ

D) defer; defer

Editing Passage: Equal Cupfuls, Different Amounts

(1) A cup measures volume, while a scale measures mass. (2) Lightly filling a cup with flour and packing it firmly can produce different masses at the same fill line. (3) Spaces between the particles affect how much flour the fixed volume contains. (4) A recipe written in grams specifies the intended amount of flour by mass. (5) Weighing that amount does not depend on the air spaces left by different filling methods. (6) A scale cannot, however, supply the writer's intended gram amount if the recipe says only "one cup." (7) Precise instructions must either specify mass or clearly describe how to measure by volume.

282. Which title best represents the whole discussion of measuring flour?

A) Why Equal Cupfuls Can Have Different Masses

B) Why Every Recipe Needs More Flour

C) How to Repair a Kitchen Scale

D) Why All Volume Measurements Are Incorrect

283. Which added sentence would most directly support the claim in sentence 2 about filling methods?

A) A cup of sugar has a different mass from a cup of flour.

B) Using one batch of flour and the same cup, a cook records 120 grams when loosely filled and 145 grams when packed to the same level.

C) A larger cup holds more flour than a smaller cup.

D) One brand of flour costs more than another brand.

284. Which version correctly distinguishes the contraction for LET US from the verb meaning ALLOWS?

A) Lets begin; this switch let’s us open the door.

B) Let’s begin; this switch let’s us open the door.

C) Lets begin; this switch lets us open the door.

D) Let’s begin; this switch lets us open the door.

285. Three discs are digital video discs, abbreviated DVD. Which sentence correctly writes the ordinary PLURAL abbreviation without possession?

A) Three DVD’s arrived in one box.

B) Three DVDs’ arrived in one box.

C) Three DVDs arrived in one box.

D) Three DVD’s’ arrived in one box.

286. Which sentence correctly distinguishes the street NAME from an ordinary direction?

A) The shop on West Street is west of the station.

B) The shop on west street is west of the station.

C) The shop on West Street is West of the station.

D) The shop on west Street is West of the station.

287. A title style capitalizes its FIRST and LAST words, even if they are short prepositions, and capitalizes other major words. Which title follows that rule?

A) The Room We Looked in

B) the Room We Looked In

C) The Room we looked In

D) The Room We Looked In

288. Choose the correctly spelled name of the musical instrument.

A) saxaphone

B) saxofone

C) saxophone

D) saxophonne

289. Which spelling names the slender, swift antelope?

A) gazelle

B) gazzelle

C) gazele

D) gazzel

290. Which spelling correctly names the garden flower?

A) dalhia

B) dahlia

C) dahila

D) dahliah

291. Which choice is a complete sentence, even though it has no direct object?

A) Because the balloon burst.

B) The balloon burst.

C) After the balloon burst.

D) The balloon bursting.

292. Which sentence correctly distinguishes the two constructions involving NUMBER?

A) A number of windows is open, but the number of broken panes is small.

B) A number of windows are open, but the number of broken panes are small.

C) A number of windows are open, but the number of broken panes is small.

D) A number of windows is open, but the number of broken panes are small.

293. Which revision preserves the meaning of "If I had known, I would have waited" while omitting IF?

A) Had I knew, I would have waited.

B) Had I known, I would have waited.

C) I had known, I would have waited.

D) Known I had, I would have waited.

294. Choose the pair: "The ___ traveler remained energetic even as the ___ rattled against the shutters."

A) hail; hale

B) hale; hale

C) hail; hail

D) hale; hail

295. Choose the pair: "The organizer will ___ the late fee and ___ to the departing visitors."

A) wave; waive

B) waive; waive

C) waive; wave

D) wave; wave

296. Choose the pair: "The mechanic will ___ the motor; the dishonest witness may ___ to hide the truth."

A) disassemble; dissemble

B) dissemble; disassemble

C) disassemble; disassemble

D) dissemble; dissemble

Editing Passage: Checking Crossword Clues

(1) Crossword editing involves several connected checks. (2) An answer must have the number of letters available in its entry. (3) A clue asking for several objects should ordinarily lead to a plural answer, while a singular clue calls for a singular one. (4) An editor must also verify that the answer really matches the clue's intended meaning; fitting the space alone is insufficient. (5) Some puzzle books have glossy covers with brightly colored titles. (6) Where entries cross, the shared square must represent the same letter in both answers. (7) These checks work together, because an entry can satisfy one requirement while failing another.

297. Which revision gives sentence 1 the most precise topic for the paragraph?

A) A crossword editor checks whether answers fit their spaces, match their clues, and agree at crossings.

B) Every attractive cover guarantees an accurate crossword.

C) The most important task is choosing the book’s cover color.

D) A crossword answer needs only the correct number of letters.

298. Which sentence should be removed because it does not develop the stated editing checks?

A) Sentence 2: An answer must have the number of letters available in its entry.

B) Sentence 4: An editor must also verify that the answer really matches the clue’s intended meaning; fitting the space alone is insufficient.

C) Sentence 6: Where entries cross, the shared square must represent the same letter in both answers.

D) Sentence 5: Some puzzle books have glossy covers with brightly colored titles.

Answer Key

Question numbers continue across all five sections.

Verbal Skills

1. D	2. C	3. D	4. A	5. C	6. A	7. B	8. B
9. A	10. A	11. B	12. C	13. B	14. D	15. C	16. B
17. C	18. B	19. A	20. B	21. B	22. C	23. A	24. D
25. C	26. A	27. C	28. D	29. D	30. D	31. C	32. A
33. C	34. D	35. D	36. C	37. C	38. D	39. A	40. D
41. A	42. B	43. B	44. A	45. B	46. B	47. C	48. D
49. D	50. A	51. B	52. C	53. B	54. B	55. D	56. D
57. C	58. C	59. A	60. A				

Quantitative Skills

61. C	62. D	63. B	64. B	65. C	66. A	67. D	68. A
69. B	70. D	71. C	72. B	73. A	74. B	75. C	76. A
77. D	78. A	79. C	80. B	81. C	82. D	83. B	84. A
85. C	86. D	87. B	88. D	89. A	90. C	91. B	92. A
93. D	94. C	95. A	96. D	97. B	98. D	99. C	100. B
101. D	102. A	103. C	104. A	105. B	106. D	107. C	108. D
109. A	110. B	111. C	112. A				

Reading Comprehension

113. B	114. D	115. C	116. C	117. A	118. A	119. A	120. C
121. D	122. B	123. D	124. B	125. A	126. D	127. B	128. C
129. C	130. C	131. D	132. A	133. B	134. B	135. C	136. B
137. A	138. B	139. D	140. A	141. D	142. D	143. C	144. A
145. A	146. C	147. D	148. C	149. D	150. B	151. D	152. C
153. B	154. B	155. B	156. D	157. A	158. A	159. A	160. C
161. A	162. A	163. D	164. C	165. C	166. D	167. C	168. B
169. D	170. D	171. B	172. A	173. B	174. B		

Mathematics

175. A	176. B	177. B	178. A	179. C	180. D	181. D	182. B
183. D	184. B	185. C	186. D	187. A	188. A	189. C	190. C
191. D	192. A	193. D	194. C	195. D	196. A	197. C	198. C
199. B	200. B	201. D	202. A	203. B	204. C	205. A	206. B

207. C	208. A	209. D	210. D	211. B	212. C	213. B	214. D
215. D	216. B	217. C	218. A	219. A	220. A	221. B	222. C
223. D	224. C	225. A	226. B	227. A	228. B	229. B	230. C
231. B	232. D	233. D	234. A	235. C	236. D	237. C	238. A

Language

239. D	240. D	241. C	242. A	243. A	244. A	245. B	246. C
247. D	248. B	249. C	250. B	251. D	252. C	253. A	254. D
255. A	256. A	257. C	258. A	259. A	260. D	261. B	262. C
263. C	264. B	265. B	266. D	267. D	268. B	269. A	270. A
271. A	272. C	273. D	274. D	275. D	276. C	277. C	278. C
279. B	280. B	281. B	282. A	283. B	284. D	285. C	286. A
287. D	288. C	289. A	290. B	291. B	292. C	293. B	294. D
295. C	296. A	297. A	298. D				

Detailed Explanations

Verbal Skills

1. HSPT-V9-001 - Answer D: extravagantly abundant
Lavish describes abundance or generosity on an extravagant scale; the unusually large selection of dishes establishes that meaning.

2. HSPT-V9-002 - Answer C: appealingly charming
Winsome means attractively charming, often in a gentle way; the audience's immediate delight supports that sense rather than authority or suspicion.

3. HSPT-V9-003 - Answer D: kindly and harmless
Benign describes a gentle, harmless quality; the intention to encourage rather than wound makes that meaning clear.

4. HSPT-V9-004 - Answer A: bustling
Desolate here describes an empty, deserted place; bustling suggests lively activity and occupied streets, reversing the stated condition.

5. HSPT-V9-005 - Answer C: lethargic
Sprightly means lively and energetic; lethargic describes sluggishness and lack of energy, whereas the other options retain the original liveliness.

6. HSPT-V9-006 - Answer A: amenable
Recalcitrant describes resistance to guidance or authority; amenable means willing to cooperate or respond to reasonable suggestions.

7. HSPT-V9-007 - Answer B: A carpenter leaves a board slightly long so it can be trimmed to fit.
Both deliberately retain extra material that can be removed during fitting; the other actions combine materials, record completion, or replace a worn part.

8. HSPT-V9-008 - Answer B: Zooming in on a digital image enlarges its display without editing the saved image.
Volume and display zoom change how stored material is presented while leaving that source intact; the alternatives alter or remove source content.

9. HSPT-V9-009 - Answer A: A conveyor carries the first packed cartons away while later cartons are still being packed.
Translation and the conveyor overlap successive stages of ongoing work; the other processes postpone the later stage until the earlier work has finished.

10. HSPT-V9-010 - Answer A: undertaking
An undertaking can be a commitment to perform an action; a prediction anticipates events without committing its speaker to make them happen.

11. HSPT-V9-011 - Answer B: memoir, autobiography, personal diary
Memoirs, autobiographies, and personal diaries concern the writer's own experiences; the other groups include works without that defining first-person life focus.

12. HSPT-V9-012 - Answer C: hierarchy
A hierarchy organizes members into higher and lower ranks; inventories, checklists, and catalogs can list items without assigning such levels.

13. HSPT-V9-013 - Answer B: Keep the parcel sealed.
Only INSPECT activates the stated exception; URGENT has no rule-changing effect, so the ordinary sealing requirement still applies.

14. HSPT-V9-014 - Answer D: Willow by Ames
Ames precedes Bell regardless of title; among the Ames reports, Willow precedes Zebra, so Willow by Ames must come first.

15. HSPT-V9-015 - Answer C: Exactly one guide spoke with visitors at the reception.
With only one guide speaking to visitors, every visitor's required conversation must involve that guide; the other conditions allow different guides for different visitors.

16. HSPT-V9-016 - Answer B: related to touch
Tactile concerns the sense of touch; following the border with fingertips supplies the contextual clue to that sensory meaning.

17. HSPT-V9-017 - Answer C: gradually diminish
Dwindle means to decrease gradually in amount or strength; the continuing dry season supports a diminishing stream rather than an abrupt flood.

18. HSPT-V9-018 - Answer B: understand fully
Fathom here means to understand something that requires thought; Luis has read the directions but still does not grasp the step's purpose.

19. HSPT-V9-019 - Answer A: gullible
Shrewd describes sharp practical judgment; gullible means easily deceived, reversing the negotiator's ability to recognize a misleading offer.

20. HSPT-V9-020 - Answer B: inarticulate
Eloquent describes effective, fluent expression; inarticulate describes difficulty expressing thoughts clearly, unlike the other choices, which suggest successful expression.

21. HSPT-V9-021 - Answer B: premeditated
Spontaneous means arising without advance planning; premeditated indicates prior consideration and planning, while the other choices preserve the absence of preparation.

22. HSPT-V9-022 - Answer C: A dispatcher gives trains traveling in opposite directions separate turns on a single-track section.
Both alternate permission to use a shared space so conflicting movements do not occur together; cleaning, labeling, and checking fares address different needs.

23. HSPT-V9-023 - Answer A: Training wheels stabilize a bicycle while its rider is learning to balance, then can be removed.
Scaffolding and training wheels provide support during a developing activity and can later be removed; the alternatives supply permanent support, restriction, or decoration.

24. HSPT-V9-024 - Answer D: A total word count can reveal that a document is too short without identifying the missing sentence.
Both summary counts can detect a discrepancy while losing the detail needed to locate it; the other checks retain information about particular items.

25. HSPT-V9-025 - Answer C: steaming
Steaming transfers heat through water vapor; broiling, toasting, and grilling ordinarily rely on dry heat rather than that shared water-based method.

26. HSPT-V9-026 - Answer A: preliminary
Preliminary describes an early, preparatory stage; the other options indicate decisions or results that have reached a settled or final condition.

27. HSPT-V9-027 - Answer C: impediment
An impediment is the obstacle itself; the other expressions, like detour, bypass, and workaround, name ways of avoiding or getting around one.

28. HSPT-V9-028 - Answer D: Rowan is immediately behind Suri.
No person stood between the adjacent pair to begin with; removing a different person while preserving order cannot insert anyone between them.

29. HSPT-V9-029 - Answer D: The last finisher in the group did not finish before any other member of the group.
A finite finishing order has a last member; that runner supplies a necessary counterexample to the claim that every member precedes another.

30. HSPT-V9-030 - Answer D: They do not establish that different people signed the pages.
Different role names need not denote different people; without information separating their identities, both descriptions could refer to the same person.

31. HSPT-V9-031 - Answer C: skillful
Deft describes skillful, coordinated handling; adjusting the small clasp without damaging its pin demonstrates that precision.

32. HSPT-V9-032 - Answer A: difficult to direct
Wayward describes something that strays from its intended course or resists direction; the wandering thoughts do not stay on the assigned task.

33. HSPT-V9-033 - Answer C: disorderly
Turbulent can describe an agitated, disorderly situation; the interruptions and arguments establish that figurative sense rather than the meeting’s length or privacy.

34. HSPT-V9-034 - Answer D: discourteous
Gracious here describes courteous, considerate behavior; discourteous reverses that quality, while the other choices preserve the original polite manner.

35. HSPT-V9-035 - Answer D: rigid
Pliable means easily bent or shaped; rigid describes resistance to bending and therefore reverses the physical quality shown by the strip.

36. HSPT-V9-036 - Answer C: ornate
Austere here means severely plain and without decoration; ornate describes elaborate decoration, reversing the chamber’s stated visual character.

37. HSPT-V9-037 - Answer C: A knot at the end of a cord keeps the cord from slipping completely through a hole.
The stop and knot act as physical barriers at an endpoint; the other devices redirect force, improve comfort, or alter reflection.

38. HSPT-V9-038 - Answer D: A breakaway link is made weaker than the main chain so that the link gives way first.
Both deliberately place a weak region where separation should occur; accidental damage, general strengthening, and locking do not provide that controlled failure point.

39. HSPT-V9-039 - Answer A: Whether two words rhyme depends on a match between their ending sounds.
Fit and rhyme are matching relationships between paired items; mass, color, and torn condition can describe an individual item without a matching partner.

40. HSPT-V9-040 - Answer D: furnace
A furnace maintains a fire or intense heat, like the listed structures; silos, cisterns, and granaries primarily store materials rather than maintain fire.

41. HSPT-V9-041 - Answer A: plateau
A plateau describes a level or period of little further change; the other expressions, like the listed words, indicate an upward change.

42. HSPT-V9-042 - Answer B: a stamp; stamp the envelope
Stamp names an object and also the action of using it; singer and sing differ in form, while red and ran retain their grammatical roles.

43. HSPT-V9-043 - Answer B: Write the words Close the shutter.
The task requires reproducing the quoted words as text and expressly distinguishes that copying from performing the action described by them.

44. HSPT-V9-044 - Answer A: The blue banner was already on that wall at noon and has not changed color.
A banner already blue on that wall at noon would contradict the noon report; later additions or removals need not contradict a past snapshot.

45. HSPT-V9-045 - Answer B: morning and busy
The shared LUM means quiet, so DAX means harbor and VEV means morning; the third example then identifies ZEP as busy.

46. HSPT-V9-046 - Answer B: joking
Jocular describes a joking or playful manner; the laughter and easing of tension show that meaning rather than seriousness or accusation.

47. HSPT-V9-047 - Answer C: disrespectfully casual
Flippant describes an inappropriately casual or dismissive attitude toward something serious; the answer’s lack of appropriate concern supplies that meaning.

48. HSPT-V9-048 - Answer D: wise and perceptive
Sagacious means showing keen judgment and wisdom; recognizing an overlooked consequence demonstrates perception rather than fame, silence, or impatience.

49. HSPT-V9-049 - Answer D: slack
Taut describes something stretched tight; slack describes looseness or a lack of tension, directly reversing the rope’s stated condition.

50. HSPT-V9-050 - Answer A: enduring
Transient means lasting only a short time; enduring means continuing or lasting, while the other options preserve the original temporary quality.

51. HSPT-V9-051 - Answer B: assertive
Meek here describes submissiveness or unwillingness to resist; assertive indicates a readiness to express and defend one’s position.

52. HSPT-V9-052 - Answer C: Caulking closes gaps around a window frame.
The gasket and caulking close gaps against unwanted passage; carrying, movement, and identification are different functions from sealing.

53. HSPT-V9-053 - Answer B: A pale border makes dark lettering easier to distinguish.
Both arrangements distinguish a feature through contrast with its surroundings; the other choices protect, move, or collect objects without that relationship.

54. HSPT-V9-054 - Answer B: One unequal pair disproves that all rods have equal length, but one equal pair does not establish it.
A universal agreement claim fails with one counterexample, while agreement in one case cannot establish every case; both comparisons use that asymmetry.

55. HSPT-V9-055 - Answer D: midnight
Midnight names a time of day like dawn, dusk, and noon; the alternatives describe a visible boundary, an instrument, or atmospheric conditions.

56. HSPT-V9-056 - Answer D: texture
Texture concerns how a surface feels or is structured; odor, perfume, and bouquet can name smells, like the listed words.

57. HSPT-V9-057 - Answer C: sheath, case, sleeve
Sheaths, cases, and sleeves can enclose and protect objects; the other groups describe cutting features, sounds, or mechanisms of movement.

58. HSPT-V9-058 - Answer C: A clear photograph of an original document
The rule distinguishes originals from copies, not clear from unclear writing; the photograph remains a copy regardless of its sharpness.

59. HSPT-V9-059 - Answer A: The rule does not determine whether the solid route is a footpath.
The statement makes dotted routes footpaths but does not require every footpath to be dotted; a solid route therefore remains unclassified by this rule.

60. HSPT-V9-060 - Answer A: At least one ROUND stamp and at least one SQUARE stamp appear.
With only two possible labels, disagreement requires both labels to occur; it identifies neither a failed machine nor any particular machine’s label.

Quantitative Skills

61. HSPT-Q9-001 - Answer C: 2,688
Since 192 = 64 × 3, its prime factorization has six factors of 2 and one of 3; it has 7 × 2 = 14 divisors, so 192 × 14 = 2,688.

62. HSPT-Q9-002 - Answer D: 2, 2
For n = 7, the remainders are 1 and 1; for n = 8, they are 0 and 2, giving outputs 2 and 2.

63. HSPT-Q9-003 - Answer B: 11/18
The numerator becomes 11, and the denominator becomes 7 + 11 = 18; both parts are updated from the preceding fraction.

64. HSPT-Q9-004 - Answer B: 101
After 100, the final digit changes from 0 to 1; no carry is needed, so the next counter reading is 101.

65. HSPT-Q9-005 - Answer C: 3/8
Let the fraction be n/d: d = 2n + 2 and d = 3n − 1, so n = 3 and d = 8.

66. HSPT-Q9-006 - Answer A: 2,318
The possible unordered pairs are 13 and 86, 16 and 83, 31 and 68, and 38 and 61; the greatest product is 38 × 61 = 2,318.

67. HSPT-Q9-007 - Answer D: 3
Taking n = 1 requires 18 to divide 12k, so k must contain a factor of 3; k = 3 also works for every n.

68. HSPT-Q9-008 - Answer A: Quantity A is greater.
Every empty seat lies between consecutive occupied seats; the two occupied endpoints make the occupied count exactly one greater.

69. HSPT-Q9-009 - Answer B: Quantity B is greater.
Their product is $(n - 1)n(n + 1) = n^3 - n$, which is smaller than n^3 because n is positive.

70. HSPT-Q9-010 - Answer D: The relationship cannot be determined from the information given.
With y = 6, x = 1 makes Quantity B greater, while x = 5 makes Quantity A greater; both choices satisfy the conditions.

71. HSPT-Q9-011 - Answer C: The two quantities are equal.
The unrolled rectangle has sides 2πr and h, so its perimeter is 2(2πr + h) = 4πr + 2h.

72. HSPT-Q9-012 - Answer B: Quantity B is greater.
Each side joining two midpoints is half the parallel original side; all three lengths, and therefore the perimeter, are halved.

73. HSPT-Q9-013 - Answer A: Quantity A is greater.
For n ≥ 5, the interior-angle sum is (n − 2) × 180° ≥ 540°, whereas the stated exterior angles total 360°.

74. HSPT-Q9-014 - Answer B: 21234
The current term 2123 contains four digits, so appending 4 produces 21234; the appended number measures the current term's length.

75. HSPT-Q9-015 - Answer C: 23, 25
After 19, discard 20, 21, 22, and 24 because of divisibility by 2 or 3; 23 and 25 both qualify.

76. HSPT-Q9-016 - Answer A: 7, 2
The next rule is median: the median of 7, 7, 5 is 7; the following range of 7, 5, 7 is 7 − 5 = 2.

77. HSPT-Q9-017 - Answer D: Together costs $0.05 more.
Separately, each $1.02 rounds to $1.00, totaling $2.00; together, $2.04 rounds to $2.05, which is five cents more.

78. HSPT-Q9-018 - Answer A: −3
Swapping the inputs preserves a + b but reverses the denominator's sign, so the result changes from 3 to −3.

79. HSPT-Q9-019 - Answer C: (3, 6)
The last two reciprocals sum to 1/2; a ≥ 4 would make their sum less than 1/2, so a = 3 and 1/b = 1/6.

80. HSPT-Q9-020 - Answer B: 14
There are 10 multiples of 6, 6 of 10, and 4 of 15; the 2 multiples of 30 were counted three times and must be excluded, giving 20 − 6 = 14.

81. HSPT-Q9-021 - Answer C: The two quantities are equal.
Place value gives N = 10a + b; subtracting the digit sum leaves 10a + b − a − b = 9a.

82. HSPT-Q9-022 - Answer D: The relationship cannot be determined from the information given.
For 2/4, the common factor 2 exceeds the reduced numerator 1; for 6/8, the common factor 2 is below the reduced numerator 3.

83. HSPT-Q9-023 - Answer B: Quantity B is greater.
Quantity A simplifies to 1/(n(n + 1)); both denominators are positive, and n(n + 1) is larger than n(n − 1), making Quantity A smaller.

84. HSPT-Q9-024 - Answer A: Quantity A is greater.
Three distinct points on a circle cannot be collinear, so they form a nondegenerate triangle; the two-side path is strictly longer than the third side.

85. HSPT-Q9-025 - Answer C: The two quantities are equal.
The equal opposite bases AB and CD have heights from P that sum to the rectangle's height; their triangle areas therefore total half its area.

86. HSPT-Q9-026 - Answer D: The relationship cannot be determined from the information given.
Without interior overlap the quantities are equal; with interior overlap Quantity A is smaller, so the information does not determine one relationship.

87. HSPT-Q9-027 - Answer B: 4, 4
Half of 8 gives 4; then half of 4 gives 2, whose smallest qualifying multiple of 4 is 4 again.

88. HSPT-Q9-028 - Answer D: 5, 5
Term 9 comes from 6 + 9 = 15, leaving 5; term 10 comes from 5 + 10 = 15, again leaving 5.

89. HSPT-Q9-029 - Answer A: 434
Moving the first digit of 343 to the end gives 433; adding 1 gives 434, so both stated operations are required.

90. HSPT-Q9-030 - Answer C: 132114
The groups in 3114 are one 3, two 1s, and one 4; writing 13, 21, and 14 in order gives 132114.

91. HSPT-Q9-031 - Answer B: −4.6
The input −4.6 rounds to −5, whose square is 25; the other inputs round to 4, −6, and 2.

92. HSPT-Q9-032 - Answer A: 6:5
Only the first ratio term is tripled, changing 2:5 to 6:5; multiplying both terms would incorrectly preserve the original ratio.

93. HSPT-Q9-033 - Answer D: 154
Their even sum requires the prime 2; the remaining primes sum to 18, allowing 5 and 13 or 7 and 11, with maximum 2 × 7 × 11 = 154.

94. HSPT-Q9-034 - Answer C: The two quantities are equal.
Each sequence requires one head and two tails, giving $p(1 - p)^2$ in either order; independence makes the products equal.

95. HSPT-Q9-035 - Answer A: Quantity A is greater.
The difference is xy + 1 − x − y = (x − 1)(y − 1); both factors are positive, making Quantity A greater.

96. HSPT-Q9-036 - Answer D: The relationship cannot be determined from the information given.
The data 1, 1, 9 give a mean above the median; 1, 9, 9 give a mean below it. Both have range 8.

97. HSPT-Q9-037 - Answer B: Quantity B is greater.
Choose a base endpoint different from the altitude’s foot: its connecting side is a right triangle’s hypotenuse, longer than the altitude and no longer than the triangle’s longest side.

98. HSPT-Q9-038 - Answer D: The relationship cannot be determined from the information given.
An inscribed square has side $r\sqrt{2}$, greater than r; an inscribed regular hexagon has side r, so the comparison can be greater or equal.

99. HSPT-Q9-039 - Answer C: The two quantities are equal.
Cutting and rearranging preserve the total amount of paper; without stretching, gaps, overlaps, or discarded pieces, the covered areas are equal.

100. HSPT-Q9-040 - Answer B: 67/27
Dividing 40/9 by 3 gives 40/27; adding 1 gives 40/27 + 27/27 = 67/27.

101. HSPT-Q9-041 - Answer D: 27
The next term uses the last two values: 11 + 17 − 1 = 27; subtracting one is part of every update.

102. HSPT-Q9-042 - Answer A: 749
The seventh input is 7 and its square is 49; concatenating those two written values gives 749 rather than their sum or product.

103. HSPT-Q9-043 - Answer C: 10
Undo the final division first: 15 × 4 = 60; then undo multiplication by 6 to obtain 60 ÷ 6 = 10.

104. HSPT-Q9-044 - Answer A: Four: 274, 472, 724, and 742
The final digit can be 2 or 4; for each choice, the other two digits can appear in either order, giving 2 × 2 = 4.

105. HSPT-Q9-045 - Answer B: 8
Write a = 5k and b = 7k; then 12k = 48 gives k = 4, so b − a = 2k = 8.

106. HSPT-Q9-046 - Answer D: 21
The full factor list of 21 is 1, 3, 7, 21, leaving exactly two factors after the endpoints are excluded.

107. HSPT-Q9-047 - Answer C: The two quantities are equal.
They are 2n + 1, 2n + 3, 2n + 5, 2n + 7, and 2n + 9, giving exactly five.

108. HSPT-Q9-048 - Answer D: The relationship cannot be determined from the information given.
For n = 6, Quantity A is 1; for n = 72, it is 12. Both satisfy divisibility, but the comparisons reverse.

109. HSPT-Q9-049 - Answer A: Quantity A is greater.
The denominator 1 − b is positive and less than 1; dividing positive a by it increases the value.

110. HSPT-Q9-050 - Answer B: Quantity B is greater.
The gap along the line of centers is 9 − 2 − 3 = 4 units, less than 5 units.

111. HSPT-Q9-051 - Answer C: The two quantities are equal.
Reflection across the symmetry plane matches the two pieces while preserving dimensions and volume, so neither piece has a greater volume.

112. HSPT-Q9-052 - Answer A: Quantity A is greater.
A quarter of the circumference is 2πr/4 = πr/2; since π/2 exceeds 1, that arc is longer than r.

Reading Comprehension

113. HSPT-R9-001 - Answer B: His coat is the same color as her uncle's, and his arm is raised.
The opening links the familiar coat color with the raised arm; no call, bag, or agreed cargo signal prompts her response.

114. HSPT-R9-002 - Answer D: She wants her mistaken wave to look like an intentional exercise.
The phrase as if describes an invented explanation for her gesture, strongly suggesting that she is trying to conceal her embarrassment.

115. HSPT-R9-003 - Answer C: examined carefully
Inspected means examined carefully; Mira's focused looking at the timetable establishes that sense, rather than rearranging, copying, or hiding it.

116. HSPT-R9-004 - Answer C: It humorously exaggerates the amount of stretching Mira does to cover her mistake.
The comparison magnifies Mira's unnecessary stretching for comic effect; nothing else establishes a real race or training session.

117. HSPT-R9-005 - Answer A: Confident greeting, embarrassed disguise, then avoidance of a full explanation
She first waves confidently, then disguises the mistake by stretching and studying the timetable, and finally avoids explaining it to her uncle.

118. HSPT-R9-006 - Answer A: It crosses the boundary at an angle rather than straight on.
The explanation explicitly links the change in direction to crossing at an angle and contrasts this with crossing straight on.

119. HSPT-R9-007 - Answer A: A change in the path of light can alter an object's apparent shape without changing the object.
The passage follows the apparent bend to its optical cause and verifies that the straw remains straight; it does not reject all visual evidence.

120. HSPT-R9-008 - Answer C: A speed change alone is insufficient to guarantee a change in direction.
The passage specifies that speed changes during straight-on crossing while direction does not; that case rules out speed change as a sufficient condition.

121. HSPT-R9-009 - Answer D: perceived rather than necessarily actual
Apparent distinguishes the position the viewer perceives from the actual position of the submerged straw; no deliberate invention or damage is involved.

122. HSPT-R9-010 - Answer B: It supplies a check on the explanation by separating a lasting physical change from a viewing effect.
The straight straw outside the water checks whether the apparent bend involved physical deformation, connecting the observation back to the proposed explanation.

123. HSPT-R9-011 - Answer D: Its sharper raised lines make a difficult design easier to examine by handling and changing the light.
The first paragraph describes handling and lighting the sharper replica to recognize a figure, not using its modern material as evidence of ancient production.

124. HSPT-R9-012 - Answer B: Its intended purpose and any uncertainty in the reconstruction
The third paragraph says the label should identify the copy's job and indicate where its reconstruction involves uncertainty.

125. HSPT-R9-013 - Answer A: They can serve valuable purposes when their abilities and historical limits are clearly distinguished.
The author defends copies for handling and interpretation while distinguishing these uses from evidence supplied by an original object's actual history.

126. HSPT-R9-014 - Answer D: The reproduced scratch shows a modern choice to imitate the mark, not the event that scratched the original.
The passage distinguishes marks resulting from an original's history from marks a modern maker chooses to reproduce, even when the appearance closely matches.

127. HSPT-R9-015 - Answer B: an attempted recreation of an earlier form
Reconstruction refers to recreating an earlier form from available evidence; the warning about uncertainty shows that some recreated features may be inferred.

128. HSPT-R9-016 - Answer C: Both can help someone understand an earlier thing or event without becoming that historical original.
The analogy clarifies that present usefulness and resemblance do not confer an earlier identity; a rehearsal can illuminate action without becoming its past performance.

129. HSPT-R9-017 - Answer C: The side of his hand passes through ink that is still wet.
The second paragraph places the blue streak where Sami's left hand has crossed wet ink, explicitly identifying the cause of the mark.

130. HSPT-R9-018 - Answer C: Effective guidance requires observing another person's actions rather than assuming one's own familiar movements will work for everyone.
The narrator moves from repeating a familiar demonstration to watching Sami's different movement and reconsidering instructions that had treated that movement as obvious.

131. HSPT-R9-019 - Answer D: It recognizes Sami's useful discovery and makes the narrator willing to learn from him.
Instead of imposing another demonstration, the narrator asks to observe Sami's successful adjustment, reversing the earlier assumption that knowledge travels only from teacher to learner.

132. HSPT-R9-020 - Answer A: a long, narrow mark
The wet ink leaves a visible blue trace on paper, so streak names a narrow mark rather than luck, repeated success, or rapid movement.

133. HSPT-R9-021 - Answer B: The visible revision gives a concrete form to the narrator's changed attitude about what a learner should find obvious.
Removing Just and naturally embodies the rejection of earlier assumptions; the remaining space suggests that more attentive instructions can now be written.

134. HSPT-R9-022 - Answer B: Above or below it rather than through it
The second paragraph states that the tilted orbit usually carries the full Moon above or below the shadow's central region.

135. HSPT-R9-023 - Answer C: Why a full Moon can occur without the precise spatial alignment needed for a lunar eclipse
The passage distinguishes the visible full-Moon phase from passage through Earth's shadow, then explains how orbital tilt separates those events.

136. HSPT-R9-024 - Answer B: Lunar eclipses would occur at every full Moon under those stated conditions.
Removing the usual above-or-below passage and explicitly placing the Moon in the shadow supplies the condition that the passage identifies as producing an eclipse.

137. HSPT-R9-025 - Answer A: It distinguishes two events, poses the apparent puzzle, and resolves it through orbital tilt and spatial alignment.
The opening separates full Moon from eclipse; the monthly puzzle then leads to an explanation involving tilt, supported by a spatial comparison.

138. HSPT-R9-026 - Answer B: Apparent alignment in a flat view does not ensure that objects occupy the same level in space.
The overpass makes a hidden third dimension familiar, clarifying why a flat drawing can conceal the separation important to the eclipse explanation.

139. HSPT-R9-027 - Answer D: They are held in positions within the crystal structure rather than free to travel through it.
The passage distinguishes local vibration from travel through the material and identifies the solid structure as the restriction on that travel.

140. HSPT-R9-028 - Answer A: Salt supplies mobile ions, whereas dissolved sugar generally consists of neutral molecules.
The sugar paragraph explicitly contrasts its neutral dissolved molecules with the mobile ions released when salt dissolves.

141. HSPT-R9-029 - Answer D: Charge must be available in particles that can move through the material; charge being present is insufficient.
All three examples separate charge from mobility and dissolution, emphasizing that charged carriers must be able to travel for the described current to flow.

142. HSPT-R9-030 - Answer D: Both make existing ions able to move beyond their fixed places in the solid structure.
The melting paragraph identifies the same release from a rigid arrangement that dissolution provides, so both examples support the importance of ion mobility.

143. HSPT-R9-031 - Answer C: able to move from place to place
Mobile describes ions freed from fixed positions and able to travel through the solution; it does not describe visibility or the absence of charge.

144. HSPT-R9-032 - Answer A: It establishes an apparent puzzle that is resolved by distinguishing the presence of charge from the ability of charges to travel.
The opening prevents a simple charge-versus-no-charge explanation; the later discussion resolves the puzzle by adding mobility as the missing condition.

145. HSPT-R9-033 - Answer A: A maintenance worker places an out-of-service notice on its doors.
The maintenance worker posts the notice before the waiting and stairwell activity begin; the passage does not attribute the stoppage to the passenger or plants.

146. HSPT-R9-034 - Answer C: Being useful to others is different from controlling their choices or being the sole connection between them.
The narrator remains useful after recognizing that neighbors connect independently and that a destination button expresses a passenger's request rather than the elevator's command.

147. HSPT-R9-035 - Answer D: They show that people maintain connections between floors even while the elevator is unavailable.
The greetings and conversation continue across floors without the elevator's help, contradicting its claim to be what keeps those floors socially connected.

148. HSPT-R9-036 - Answer C: The elevator describes feeling important, being surprised, and revising its own beliefs.
Giving an elevator feelings, beliefs, and self-reflection attributes human mental qualities to a nonhuman object; the other details are ordinary setting information.

149. HSPT-R9-037 - Answer D: Self-importance gives way to surprise, then to a more modest satisfaction in useful work.
The narrator first enjoys being missed, then hears independent connections between neighbors, and finally accepts its useful role without claiming authority over the building.

150. HSPT-R9-038 - Answer B: A gray curtain of rain beyond the fence
The passage describes a gray curtain hanging below the dark band, and Niko identifies rain falling there before it reaches the house.

151. HSPT-R9-039 - Answer D: He begins to value a feature of the unfamiliar landscape that his first comparison with home had made him overlook.
The open distance first seems empty because it lacks familiar mountains; watching the approaching rain reveals a view his home landscape would hide.

152. HSPT-R9-040 - Answer C: He now expects the open view to offer something worth noticing.
His earlier complaint about having nothing to look at contrasts with his final choice to keep watching, supporting a changed expectation rather than forgetfulness.

153. HSPT-R9-041 - Answer B: a narrow gap
Niko looks between two peaks through a familiar opening in the skyline, so notch means a narrow gap rather than a forecast, wind, or improvement.

154. HSPT-R9-042 - Answer B: A hanging, sheetlike area of rain that gradually widens across Niko's view
The curtain comparison gives the rain a visible hanging shape; it describes the appearance of distant rainfall rather than an actual piece of fabric.

155. HSPT-R9-043 - Answer B: Corresponding upper-arm and forearm bones
The second paragraph names corresponding upper-arm and forearm bones while explicitly allowing their proportions and functions to differ.

156. HSPT-R9-044 - Answer D: A grouping criterion should fit the question being asked; similar function and similar underlying structure are different criteria.
The passage presents function and underlying anatomy as useful for different purposes, then advises readers to match a grouping rule to its intended question.

157. HSPT-R9-045 - Answer A: Its different job would not prevent a structural comparison with a human arm or bird's wing.
The passage allows corresponding structures to serve different jobs; a swimming function therefore does not eliminate the described anatomical basis for comparison.

158. HSPT-R9-046 - Answer A: matching in relative position and relationship
The passage ties correspondence to positions and connections, while allowing shape, proportions, and uses to differ; matching relationships is therefore the intended sense.

159. HSPT-R9-047 - Answer A: To show that the first criterion answers a real question even though it cannot answer every anatomical or evolutionary question
The author limits what functional grouping can establish without dismissing its legitimate purpose, preparing the recommendation to choose criteria according to the question.

160. HSPT-R9-048 - Answer C: The worker says its hands were advanced during testing.
The worker explains that the hands were moved forward and not restored, undermining Sera's reliance on the displayed time.

161. HSPT-R9-049 - Answer A: It would conceal the mistaken accusation she had been preparing.
Calling that answer attractive and impossible to check shows its usefulness for concealing her earlier mistaken reaction.

162. HSPT-R9-050 - Answer A: reacted with irritation
The sharpening complaints and disappointment at mistaken sightings establish irritation, rather than a literal physical change or movement.

163. HSPT-R9-051 - Answer D: Her intended accusation has become an uncomfortable burden.
The figurative weight connects her discomfort about the accusation with the actual weight she has been carrying.

164. HSPT-R9-052 - Answer C: She prepares blame, discovers a faulty assumption, and chooses an honest admission.
The explanation overturns her assumption; she then acknowledges her reaction instead of delivering the accusation she prepared.

165. HSPT-R9-053 - Answer C: All source light is blocked in the darkest region; only some is blocked in the penumbra.
The model distinguishes complete blockage from partial blockage, where light from unblocked portions still reaches the surface.

166. HSPT-R9-054 - Answer D: Partial blockage of an extended source creates a gradual brightness transition.
The displaced-shadow model explains the partially illuminated transition, including why a sharp object can cast a soft edge.

167. HSPT-R9-055 - Answer C: The shadow alone does not establish that the card’s edge is physically fuzzy.
The passage explicitly allows lighting geometry to produce a broad transition without changing the card’s physical edge.

168. HSPT-R9-056 - Answer B: a region receiving light from only part of the source
The third paragraph defines the term through partial shading, where some source light is blocked and some arrives.

169. HSPT-R9-057 - Answer D: To distinguish contributions from different positions and explain partial blockage
The imagined points clarify how some contributions are blocked while others arrive; the model is not a literal construction claim.

170. HSPT-R9-058 - Answer D: An encounter can remain unresolved while a character’s changed attitude becomes clear.
The essay separates uncertainty about the door’s answer from the visible attitude change expressed by choosing to knock.

171. HSPT-R9-059 - Answer B: It connects the final gesture with the traveler’s earlier stated attitude.
The contrast between using the key and knocking supplies textual evidence; the ship prediction has no comparable support.

172. HSPT-R9-060 - Answer A: leave undisclosed
The ending does not reveal the encounter’s outcome, so withhold refers to leaving information undisclosed rather than disproving it.

173. HSPT-R9-061 - Answer B: Their contrast makes a change in the traveler’s attitude visible through action.
The key initially expresses assumed entitlement, while knocking makes room for a response; the contrasting actions embody the change.

174. HSPT-R9-062 - Answer B: Who answers the knock and what happens afterward
The example supplies the arrival, key, and knock, then stops before any answer, preserving uncertainty about what follows.

Mathematics

175. HSPT-M9-001 - Answer A: 0.3
The greatest integer not exceeding −2.7 is −3; subtracting it gives −2.7 − (−3) = 0.3.

176. HSPT-M9-002 - Answer B: −0.5
The first move ends at −4 + 7 = 3; the return distance is 7/2 = 3.5, so the final position is −0.5.

177. HSPT-M9-003 - Answer B: 11
The sum is a + b/10 + b + a/10 = 1.1(a + b); dividing 12.1 by 1.1 gives 11.

178. HSPT-M9-004 - Answer A: 30
Let the original total be N: 2N/3 − 5 = 3(N − 5)/5; multiplying by 15 gives 10N − 75 = 9N − 45, so N = 30.

179. HSPT-M9-005 - Answer C: 0
The expressions are 2x + 2k + 1 and 2x + 1 + k; equality for every x requires 2k = k, so k = 0.

180. HSPT-M9-006 - Answer D: −2 < x < 1
The quotient is negative only when numerator and denominator have opposite signs; that happens for x − 1 < 0 and x + 2 > 0.

181. HSPT-M9-007 - Answer D: n = 4; value = 21
For n = 1, 2, and 3, the values are 3, 7, and 13; n = 4 gives 21 = 3 × 7.

182. HSPT-M9-008 - Answer B: Every x from 2 through 7, including both endpoints
For 2 ≤ x ≤ 7, the sum is (x − 2) + (7 − x) = 5; outside this interval, the sum is greater.

183. HSPT-M9-009 - Answer D: The reflected rectangle does not coincide with the original because this rectangle is not a square.
Only a square has diagonal reflection symmetry among rectangles; reflection fixes the diagonal and preserves area, but this nonsquare rectangle will not coincide.

184. HSPT-M9-010 - Answer B: 24 + 25π/2 square centimeters
The hypotenuse is √(6² + 8²) = 10, so the radius is 5; the areas sum to 6 × 8/2 + π × 5²/2.

185. HSPT-M9-011 - Answer C: The face originally pointing north
Track the top face after each roll: originally west, then south, then down, then north; the final top face is the original north face.

186. HSPT-M9-012 - Answer D: 90 seconds
The first 20 seconds cover 20 × (1 + 0.5) = 30 meters; the remaining 70 meters take 70 seconds, for 90 seconds total.

187. HSPT-M9-013 - Answer A: 12 minutes
The 5-minute glass empties again at minute 10; the 8-minute glass has run 2 minutes since its flip, so reversing it leaves 2 minutes until empty.

188. HSPT-M9-014 - Answer A: 1/4
The qualifying tiles are 4 and 8, giving 2/8 = 1/4; the independent spinner result does not change their probability.

189. HSPT-M9-015 - Answer C: It is between 10 and 19 minutes, inclusive.
With fourteen observations, the median averages the seventh and eighth ordered values; both fall in the second group, so their average lies from 10 through 19.

190. HSPT-M9-016 - Answer C: 1/12
There are 24 equally likely orders of the physical tiles; the two O tiles can exchange places in the desired word, giving 2/24 = 1/12.

191. HSPT-M9-017 - Answer D: 3 red tokens and 2 blue tokens
Three red tokens contribute −6 and two blue tokens contribute +6; their values cancel to zero.

192. HSPT-M9-018 - Answer A: 30
The product of two positive integers equals their greatest common factor times their least common multiple, so b = 6 × 90 ÷ 18 = 30.

193. HSPT-M9-019 - Answer D: 5
Since 18n = 9 × 2n, 2n must be a square; the qualifying values n are 2, 8, 18, 32, and 50.

194. HSPT-M9-020 - Answer C: 3
The first equation gives x = 2; substituting it into the second gives 2k + 4 = 10, so k = 3.

195. HSPT-M9-021 - Answer D: n² + 2n − 8
There are n − 2 used rows, each with n + 4 wells; expanding (n − 2)(n + 4) gives n² + 2n − 8.

196. HSPT-M9-022 - Answer A: A/(2h) − d
Divide both sides by 2h to obtain w + d = A/(2h), then subtract d outside the resulting fraction.

197. HSPT-M9-023 - Answer C: x = 6 only
The first inequality gives x < 7; since x is positive, the second gives x > 5, leaving only the integer 6.

198. HSPT-M9-024 - Answer C: 8 triangular faces
Each pyramid contributes four triangular side faces; the two square bases become internal, leaving 4 + 4 = 8 exposed faces.

199. HSPT-M9-025 - Answer B: (6, 6)
The radius satisfies πr² = 36π, so r = 6; tangency to each axis places the center six units from that axis.

200. HSPT-M9-026 - Answer B: 2√3 units
The cube's space diagonal is √(4² + 4² + 4²) = 4√3; the center is its midpoint, so the sphere's radius is 2√3.

201. HSPT-M9-027 - Answer D: No; it exceeds the limit by 1 ounce.
The parts and wrapping total 13 + 17 + 3 = 33 ounces; the 2-pound limit is 32 ounces, so the parcel is one ounce over.

202. HSPT-M9-028 - Answer A: 260 milliampere-hours
Convert the intervals to 2/3 hour and 1/2 hour; the charge is 300 × 2/3 + 120 × 1/2 = 260 milliampere-hours.

203. HSPT-M9-029 - Answer B: 4.8 grams
The coated area is 2 × 25 × 40 = 2,000 square centimeters; thickness is 0.002 centimeter, giving 4 cubic centimeters and 4 × 1.2 = 4.8 grams.

204. HSPT-M9-030 - Answer C: Minute 2
The separate minute counts are 18, 47 − 18 = 29, and 65 − 47 = 18; the largest increase occurs during minute 2.

205. HSPT-M9-031 - Answer A: 5/12
Different colors occur as green-red or red-green: (3/4 × 1/3) + (1/4 × 2/3) = 1/4 + 1/6 = 5/12.

206. HSPT-M9-032 - Answer B: P has the higher acceptable-part rate in each session, but Q has the higher rate overall.
P leads 90% to 80% and 60% to 50% separately; overall P makes 45/70 acceptable parts, while Q makes 53/70, so Q leads overall.

207. HSPT-M9-033 - Answer C: 0.150
Fifteen hundredths is 0.15; a zero added at the right end of its decimal digits gives the equal value 0.150.

208. HSPT-M9-034 - Answer A: 0.70
The other settings equal 1/8, 3/8, and 5/8; 0.70 × 8 = 5.6 is not an integer number of steps.

209. HSPT-M9-035 - Answer D: 1/6
The full capacity is 6 × 4 = 24 milliliters; 4 milliliters is 4/24 = 1/6 of that whole.

210. HSPT-M9-036 - Answer D: 88
A square ending in 4 comes from a number ending in 2 or 8; above 88, candidates 92 and 98 fail, while 88² = 7,744 qualifies.

211. HSPT-M9-037 - Answer B: 17
First F(4) = 5 + 5 + 1 = 11; then F(6) = F(4) + F(2) + 1 = 11 + 5 + 1 = 17.

212. HSPT-M9-038 - Answer C: 36
The difference of squares factors as (a + b)(a − b), so its value is 9 × 4 = 36.

213. HSPT-M9-039 - Answer B: x ≥ 1 and x ≠ 4
The square root requires x − 1 ≥ 0, while the denominator requires x − 4 ≠ 0; both restrictions must hold together.

214. HSPT-M9-040 - Answer D: No set of real values satisfies all four equations.
The first and third equations give a + b + c + d = 17; the second and fourth give the same sum as 18, a contradiction.

215. HSPT-M9-041 - Answer D: (0, 5)
Every point on the y-axis has x = 0; keeping P’s height preserves its y-coordinate of 5, giving Q = (0, 5).

216. HSPT-M9-042 - Answer B: (1, 4)
Translating (1, 4) gives (3, 4); reflecting it across x = 2 also gives (3, 4), whereas the other points produce different results.

217. HSPT-M9-043 - Answer C: √3
Similarity gives L/W = W/(L/3) = 3W/L; therefore (L/W)² = 3, and the positive ratio is √3.

218. HSPT-M9-044 - Answer A: 2:15 p.m.
Clock A reads 12:15 p.m. after 25 minutes; adding the fixed two-hour difference gives Clock B’s reading of 2:15 p.m.

219. HSPT-M9-045 - Answer A: Ruler B, by 0.1 millimeter
Ruler A’s spacing is 10/20 = 0.5 millimeter; Ruler B’s 0.4-millimeter spacing is smaller by 0.1 millimeter.

220. HSPT-M9-046 - Answer A: 3/8
There are 16 equally likely records; the two changes can occupy three pairs of gaps, with two choices of starting label, giving 6/16 = 3/8.

221. HSPT-M9-047 - Answer B: Median 4; range 12
The decreasing transformation sends the median to 20 − 2 × 8 = 4; distances scale by 2, so the range becomes 12.

222. HSPT-M9-048 - Answer C: 2/3
The recorded number is 1 after original rolls of 1, 4, 5, or 6; those four equally likely outcomes give 4/6 = 2/3.

223. HSPT-M9-049 - Answer D: 0.06
Three fifths, 0.60, and 60% equal six tenths; 0.06 instead represents six hundredths, a smaller value.

224. HSPT-M9-050 - Answer C: 6
The reduced fraction must be k/7 with k from 1 through 6; multiplying by 12 gives six original numerators.

225. HSPT-M9-051 - Answer A: 3, exactly
The decimals equal 36/99 and 12/99; division cancels their common denominator and leaves 36/12 = 3 exactly.

226. HSPT-M9-052 - Answer B: x = 0 only
Expanding the left side gives $x^2 + 4x + 4$; equality with $x^2 + 4$ requires $4x = 0$.

227. HSPT-M9-053 - Answer A: 42
Multiplying the entire given equation by 6 yields $2x + 3y = 42$ without finding the individual variables.

228. HSPT-M9-054 - Answer B: x = 3
If $x \geq 5$, the equation would require $x = x + 2$; otherwise $5 = x + 2$ gives $x = 3$.

229. HSPT-M9-055 - Answer B: No real x in the stated domain satisfies it.
Valid cross multiplication gives $x^2 + x = x^2 + x - 2$, a contradiction that no permitted value can satisfy.

230. HSPT-M9-056 - Answer C: 7 units
Every route requires at least three horizontal and four vertical units; a route without backtracking achieves their sum, 7.

231. HSPT-M9-057 - Answer B: 4 units
The 6-unit base difference splits into two 3-unit offsets; a leg forms a right triangle, giving height $\sqrt{(5^2 - 3^2)} = 4$.

232. HSPT-M9-058 - Answer D: 24
Each painted face contains 16 cubes; their shared edge has 4 double-painted cubes, giving 32 − 2 × 4 = 24.

233. HSPT-M9-059 - Answer D: 21:05
For this p.m. hour, add 12 to the hour and preserve the minutes; 9 + 12 gives 21:05.

234. HSPT-M9-060 - Answer A: 136 meters
The round-trip distance is 340 × 0.8 = 272 meters; the wall is half that total distance away, or 136 meters.

235. HSPT-M9-061 - Answer C: 30 centimeters
Six cuts lose 12 millimeters, leaving 2,112 − 12 = 2,100 millimeters; dividing by seven gives 300 millimeters, or 30 centimeters.

236. HSPT-M9-062 - Answer D: 1/3
Two of the six flagged packages are correctly labeled; the false-alarm fraction among flagged packages is therefore 2/6 = 1/3.

237. HSPT-M9-063 - Answer C: 10
The old total is 48; the new total ranges from 52 to 64, so the new mean ranges from 10.4 to 12.8.

238. HSPT-M9-064 - Answer A: 3/7
Excluding TTT leaves seven equally likely records; HHT, HTH, and THH are the three with exactly two heads.

Language

239. HSPT-L9-001 - Answer D: The shell is thin, not fragile, and it bends slightly.
The contrasting interruption not fragile is set off as a unit; commas should not split that phrase or separate the subject from its verb.

240. HSPT-L9-002 - Answer D: The editor called the ending "abrupt"; the writer agreed.
A semicolon joining independent clauses goes outside quotation marks; it belongs after the complete first clause, not inside the quoted word.

241. HSPT-L9-003 - Answer C: The kit holds three essentials: a compass, a map, and a lamp; a separate pouch holds batteries.
The complete introductory clause takes a colon before its list; the semicolon then joins that expanded clause to the independent statement about batteries.

242. HSPT-L9-004 - Answer A: The Greek letter delta appears beside the diagram.
Greek is a proper adjective and is capitalized; letter and delta are ordinary nouns in this sentence and remain lowercase.

243. HSPT-L9-005 - Answer A: Crossing the Atlantic in winter
Sentence case capitalizes Crossing as the first word and Atlantic as a geographic proper noun; ordinary words, including winter, remain lowercase.

244. HSPT-L9-006 - Answer A: sherbet
Sherbet uses e in its final syllable and one final t; these alternatives alter that vowel or add an unnecessary consonant.

245. HSPT-L9-007 - Answer B: fuchsia
Fuchsia places ch before s and ends ia; each alternative rearranges or omits part of that distinctive consonant and vowel sequence.

246. HSPT-L9-008 - Answer C: climbing
Worth takes a noun or gerund complement in this construction, so climbing correctly names the activity being judged valuable.

247. HSPT-L9-009 - Answer D: had been flying
Had been flying places an ongoing action before the past reference point landed and expresses its duration up to that point.

248. HSPT-L9-010 - Answer B: Not only did the guide find the path, but the guide also explained how it had formed.
Initial not only triggers auxiliary-subject inversion; did carries the past tense, so the following main verb stays in its base form find.

249. HSPT-L9-011 - Answer C: expedient; expeditious
Expedient means suited to an immediate practical purpose; expeditious means prompt and efficient, so the patch and response require different adjectives.

250. HSPT-L9-012 - Answer B: delegated; relegated
Delegated means entrusted responsibility or authority to another person; relegated means assigned something to a lower or less important position.

251. HSPT-L9-013 - Answer D: A small arrow printed beside the ridge can show which side should face up.
The arrow adds a specific link between the protective ridge and the handling instruction; the alternatives repeat information or introduce irrelevant details.

252. HSPT-L9-014 - Answer C: To prevent this,
Sentence 3 describes a design intended to prevent the flattening problem in sentence 2; To prevent this makes that problem-solution connection explicit.

253. HSPT-L9-015 - Answer A: The maker prefers painting large outdoor walls to decorating small paper objects.
The preference for painting walls does not develop the package's protection or instructions; the other sentences explain features relevant to the stated packaging problem.

254. HSPT-L9-016 - Answer D: The bell rang at five o'clock.
The fixed expression o'clock places the apostrophe immediately after o to mark the omitted letters in the older expression of the clock.

255. HSPT-L9-017 - Answer A: Did the guide write, "The gate is closed"?
The question belongs to the complete surrounding sentence, not to the quoted statement, so its single question mark goes outside the closing quotation mark.

256. HSPT-L9-018 - Answer A: The force is measured in newtons, a unit named after Newton.
Newton is a person's proper name; the spelled-out unit name newton is a common noun and remains lowercase.

257. HSPT-L9-019 - Answer C: Hurricane Iris weakened overnight; the hurricane continued north.
Hurricane Iris is the storm's proper name; the later generic hurricane and the compass direction north are lowercase.

258. HSPT-L9-020 - Answer A: macaroni
Macaroni has one n and ends oni; the alternatives change its final vowel spelling or unnecessarily double the n.

259. HSPT-L9-021 - Answer A: tapestry
Tapestry has one p, uses e after tap, and ends try; the alternatives change that vowel, double p, or alter the ending.

260. HSPT-L9-022 - Answer D: mezzanine
Mezzanine uses double z, one n before i, and the ending ine; the alternatives omit a z, double n, or change a vowel.

261. HSPT-L9-023 - Answer B: Rowleys
A family surname ending in y keeps the y and adds s for the plural; no apostrophe is needed without possession.

262. HSPT-L9-024 - Answer C: shaped
Shaped begins a reduced passive relative clause modifying ceramics; are supplies the main verb, so another finite verb would break the sentence.

263. HSPT-L9-025 - Answer C: Whomever
Whomever is the object of selects within its clause; the entire clause serves as the subject of will guide, but that does not change the pronoun's role.

264. HSPT-L9-026 - Answer B: decent; descent; dissent
Decent means satisfactory, descent is downward movement, and dissent is disagreement; shelter, the steep route, and the group's agreement require those respective meanings.

265. HSPT-L9-027 - Answer B: borne; born
Borne is the past participle meaning carried; born is used for coming into life, so the moving boat and the builder require different forms.

266. HSPT-L9-028 - Answer D: 4, 5, 3
Sentence 4 draws the line, sentence 5 checks readability through it, and sentence 3 expressly permits adding the correction only after that check.

267. HSPT-L9-029 - Answer D: Readable corrections let later readers check the log's history without losing the original record.
The conclusion unifies preserved entries, visible corrections, and later review; the alternatives introduce unrelated decoration or contradict the described purpose.

268. HSPT-L9-030 - Answer B: The corrected information shows what changed, and the initials show who made the change.
The combination preserves the separate functions of the correction and initials; the alternatives exchange those functions, erase the distinction, or add an unsupported cause.

269. HSPT-L9-031 - Answer A: "I thought the answer was—"
An em dash signals a sudden break in speech; an ellipsis usually signals trailing off, while a colon or comma does not supply that abrupt ending.

270. HSPT-L9-032 - Answer A: the sisters-in-law’s car
The plural is sisters-in-law; because the whole compound ends in law rather than s, its possessive adds apostrophe-s at the end.

271. HSPT-L9-033 - Answer A: "[Nora] left at dawn."
Square brackets identify the editor’s replacement Nora; marking another portion or using parentheses would not identify the specified editorial change.

272. HSPT-L9-034 - Answer C: I wrote a lowercase i below the line.
The first-person pronoun I is always capitalized; the letter explicitly described as lowercase remains i, and lowercase is an ordinary adjective.

273. HSPT-L9-035 - Answer D: The chapter compares Bronze Age metalwork with modern casting.
Bronze Age names a particular historical period and takes capitals on both words; metalwork, modern, and casting are ordinary descriptive words.

274. HSPT-L9-036 - Answer D: raccoon
Raccoon has double c and double o, followed by one n; the alternatives omit a vowel, add a consonant, or change a vowel.

275. HSPT-L9-037 - Answer D: moccasin
Moccasin ends asin, with one final n; the alternatives double that n, change i to a, or replace the final n with m.

276. HSPT-L9-038 - Answer C: you
An imperative normally addresses an understood you; the towel is the direct object, and Ken is the object of the preposition to.

277. HSPT-L9-039 - Answer C: are; carries
Criteria is plural and takes are; one criterion is singular and takes carries, regardless of the nearby plural word others.

278. HSPT-L9-040 - Answer C: The final note having faded, the audience remained still.
Having faded forms a nonfinite opening phrase with its own subject; remained supplies the main verb, completing the sentence without a comma splice.

279. HSPT-L9-041 - Answer B: leek; leak
A leek is a vegetable that can be sliced; a leak is an unintended escape of liquid or gas that can be repaired.

280. HSPT-L9-042 - Answer B: bolder; boulder
Bolder is the comparative adjective meaning more courageous; a boulder is a large rock, so the two blanks require different meanings.

281. HSPT-L9-043 - Answer B: differ; defer
Differ means disagree or hold different views; defer means postpone, which fits delaying the decision until a later day.

282. HSPT-L9-044 - Answer A: Why Equal Cupfuls Can Have Different Masses
The passage explains how packing changes the mass in a fixed volume and why recipes should specify a clear measurement method, without rejecting volume altogether.

283. HSPT-L9-045 - Answer B: Using one batch of flour and the same cup, a cook records 120 grams when loosely filled and 145 grams when packed to the same level.
The comparison keeps flour, cup, and fill level constant while changing packing; the other observations change the material, capacity, or price instead.

284. HSPT-L9-046 - Answer D: Let’s begin; this switch lets us open the door.
Let’s contracts let us and needs an apostrophe; lets means allows and contains no omitted word to mark.

285. HSPT-L9-047 - Answer C: Three DVDs arrived in one box.
The plural of this abbreviation adds s without an apostrophe; no ownership or omitted letters are being expressed.

286. HSPT-L9-048 - Answer A: The shop on West Street is west of the station.
West Street is a proper street name; the later west states a direction rather than naming a place.

287. HSPT-L9-049 - Answer D: The Room We Looked In
In is capitalized because it is the last title word; The is first, and Room, We, and Looked are major words.

288. HSPT-L9-050 - Answer C: saxophone
Saxophone contains xo followed by phone; the alternatives change the internal vowel, replace ph, or add an extra n.

289. HSPT-L9-051 - Answer A: gazelle
Gazelle uses one z and double l before its final e; the alternatives alter those consonant counts or the ending.

290. HSPT-L9-052 - Answer B: dahlia
Dahlia places h directly after a and ends lia; the alternatives reverse letters or add an unnecessary final h.

291. HSPT-L9-053 - Answer B: The balloon burst.
Burst is an intransitive main verb here and needs no direct object; the alternatives lack an independent main clause.

292. HSPT-L9-054 - Answer C: A number of windows are open, but the number of broken panes is small.
A number of means several and takes a plural verb; the number identifies one total and takes a singular verb.

293. HSPT-L9-055 - Answer B: Had I known, I would have waited.
A past counterfactual conditional can omit if by placing had before the subject, while retaining the past participle known.

294. HSPT-L9-056 - Answer D: hale; hail
Hale describes robust health, while hail names frozen precipitation; the traveler’s condition and the falling ice require those different words.

295. HSPT-L9-057 - Answer C: waive; wave
Waive means relinquish or choose not to enforce; wave means make the hand gesture, so the fee and visitors require different verbs.

296. HSPT-L9-058 - Answer A: disassemble; dissemble
Disassemble means take apart; dissemble means conceal one’s true motives or feelings through pretense, which fits the witness’s attempt to hide the truth.

297. HSPT-L9-059 - Answer A: A crossword editor checks whether answers fit their spaces, match their clues, and agree at crossings.
The revision includes the paragraph’s linked tests of length, meaning, grammatical number, and crossing consistency without reducing editing to one check.

298. HSPT-L9-060 - Answer D: Sentence 5: Some puzzle books have glossy covers with brightly colored titles.
Sentence 5 concerns cover appearance, whereas the other sentences explain constraints on the fit, meaning, or shared letters of entries.